Certified Feature Selection for Robust Hypothesis Testing

An experimental study of global search and independent statistical utility

Protocol frozen 3 October 2026 · Research report

1 One question, two kinds of evidence

We want to choose a small set of measurements that makes two uncertain class distributions distinguishable. The proposed selector minimizes a two-class Wasserstein robust testing risk under a feature budget. The theory explains how a continuous test problem can supply global lower cuts for the discrete selection problem. This report asks whether that construction is computationally useful and whether the resulting measurements are useful on new data.

These are separate questions. An accurate global certificate establishes how well a particular empirical objective has been optimized. It does not establish that the objective is the best selection principle for a population, that the chosen radius covers that population, or that a two-sample test has high power. Conversely, a selector can perform well on held-out observations without providing a global certificate for its training criterion.

We therefore organize the study around two evidence lines. The *computational line* checks the finite formulation and its cuts, then compares original-objective upper bounds, global lower bounds, time, search effort, and unsuccessful runs. The *statistical line* uses independent samples to assess equality-test calibration and power, signal recovery, prediction, and response to specified distribution shifts. A common downstream evaluation prevents differences in prediction architecture from being mistaken for differences in selected coordinates.

Table 1: Recorded outcome inventory. Complete denotes completed evaluation, not certified optimal selection.

Suite	Planned	Present	Evaluated
statistical	960	960	960
computational	320	320	320
real	240	240	240
sensitivity	60	60	60
ratio	60	60	60

The saved verification run reports 106 passed tests in 8.72 seconds. The host is AMD Ryzen 9 7945HX with Radeon Graphics. Summed charged selection time is 2.94 hours; summed end-to-end trial time, including evaluation, is 2.96 hours. These sums include unsuccessful outcomes and recorded remediation costs; they are not elapsed calendar time for concurrent runners.

The LP and MILP engine is SciPy-backed HiGHS 1.12.0 (SciPy 1.18.1). Logistic correctness checks use CVXPY 1.9.3 with Clarabel 0.11.1. An independently installed highspy package is not used to identify SciPy's bundled engine.

Build revision: cbc2e5e7b7be. Input and configuration SHA-256 hashes are saved with the generated report manifest.

The completed small-instance comparisons contain 159 paired, numerically solved gate/reference outcomes. Their largest absolute feasible-objective difference is 0.000; 0 recorded gate intervals fail the paired enu-

meration compatibility check at the combined declared tolerances. Agreement validates the shared integer target, rather than a statistical performance gain.

The primary comparisons do not establish better statistical selection from the robust objective: mean, variance, and dependence results expose weak recovery or power relative to other selectors. The stronger risk-scaled gate is also not uniformly faster in the current master implementation. A separately declared exploratory supplement retains the full dual fraction to address the observed computational limitation. The main results and this supplement are reported separately.

1.1 Why this evaluation follows the local literature

The experimental sections and relevant appendices of the source papers were read before the protocol and implementation. Wang, Dey, and Xie [4] motivate independent selection and final testing, power paired with null rejection, and false-discovery/non-discovery proportions. Xie, Gao, and Xie [2] motivate small training samples, nonlinear distribution differences, and independent classification error. Gao et al. [3] motivate inspecting uncertainty radii. Swaroop, Krishnamurti, and Wilder [6] motivate a separate downstream-training split and a common prediction pipeline. Bertsimas, Cory-Wright, and Pauphilet [1] motivate computational ablations and explicit time-limited outcomes.

We transfer these evaluation ideas to our task. We do not identify the two class labels with the multiple deployment populations in Swaroop et al., or relabel sequential detection delay as fixed-sample testing power. Our synthetic mechanisms and sample sizes are specified adaptations. The baseline section records where a mathematical reconstruction differs from inspected author code.

2 What the algorithm does, from one support to global search

2.1 The original testing target

Class $k \in \{0, 1\}$ supplies observations $x_{ki} \in \mathbb{R}^D$ with positive weights q_{ki} summing to one within that class. Its empirical measure is $\widehat{P}_k = \sum_i q_{ki}\delta_{x_{ki}}$. The true class distribution is protected only if it belongs to the stated ball $W_1(P_k, \widehat{P}_k) \le \theta_k$, with strictly positive radius θ_k. The ground cost is L1 on the full standardized observation space.

A binary mask z selects coordinates $S(z) = \{j : z_j = 1\}$. A positive score g favors class zero. The class-zero and class-one losses are respectively $\Phi(-g)$ and $\Phi(g)$. Our primary loss is half hinge, $\Phi(v) = (1 + v)_+/2$; logistic loss is used in secondary correctness checks. The objective is

$$\mathrm{OPT} = \min_{\substack{z\in\{0,1\}^D \\ \mathbf{1}^\top z \le d}} V(z), \qquad V(z) = \inf_{g=g(x_{S(z)})} \left[\sup_{P_0} \mathbb{E}_{P_0}\Phi(-g) + \sup_{P_1} \mathbb{E}_{P_1}\Phi(g) \right]. \tag{1}$$

The two suprema use their respective Wasserstein balls. A test may ignore an additional observed coordinate, so an optimal support can be padded to exactly d coordinates. This justifies the full-budget equality in the implemented master. It does not establish that every selected variable has unique population relevance.

Half-hinge scores can be clipped to $[-1, 1]$. Writing $\pi_0 = (1 + g)/2$, the losses become $1 - \pi_0$ and π_0, the two randomized error probabilities. Thus V is a *sum* of classwise risks; its equal-prior balanced average is $V/2$. The uninformative test has risk $B = 1$. Logistic has the corresponding constant-test risk $2 \log 2$.

2.2 One fixed support becomes a convex test problem

Pool all source observations as targets $t_1, \dots, t_N$. Let $\Delta_{kimj} = |x_{ki,j} - t_{m,j}|$. The fixed-support oracle solves

$$\begin{aligned} V(z) = \min_{a,u,\lambda} \quad & J(a,\lambda) := \sum_{ki} q_{ki} a_{ki} + \sum_{k} \theta_k \lambda_k \\ \text{subject to} \quad & a_{ki} + \lambda_k \sum_{j} z_j \Delta_{kimj} \geq \Phi(\sigma_k u_m), \qquad \lambda_k \geq 0, \end{aligned} \tag{2}$$

where $\sigma_0 = -1$, $\sigma_1 = 1$. The score u_m is the test value at target m. The potential a_{ki} bounds a source atom's loss, and λ_k prices additional transport. The potentials and scores are free variables; self-target constraints make feasible potentials nonnegative. For half hinge, using $r_m = (1 + u_m)/2 \in [0, 1]$ makes this an ordinary linear program.

The finite pool is exact for the stated two losses and ambient-space assumptions. The theory note proves this by finite transport-LP duality in one direction and a constructive extension to every observation in the other. In particular, the test is not deployed by an arbitrary interpolation of pool scores. It uses compatible whole-space loss envelopes,

$$E_k(x) = \min_i \left\{ a_{ki} + \lambda_k \sum_{j \in S(z)} |x_{ki,j} - x_j| \right\}. \tag{3}$$

For half hinge, take the midpoint of $[\max(0, 1 - E_0(x)), \min(1, E_1(x))]$ as $\pi_0(x)$. The interval is nonempty by the finite constraints and the triangle inequality. Integrating these envelope bounds over any admissible transport gives classwise risk at most $q_k^\top a_k + \theta_k \lambda_k$. A repaired primal solution therefore supplies a feasible-test *upper* bound.

2.3 Why a support evaluation can inform every other support

Introduce a two-class coordinate-price block $w_j = (w_{0j}, w_{1j})$. An inactive coordinate forces $w_j = 0$; an active coordinate allows $0 \leq w_{kj} \leq \lambda_k$. Since distance entries are nonnegative, saturating an active price to λ_k preserves loss feasibility. The logical-coordinate formulation therefore has the same binary optimum as (2).

Nonnegative potentials imply $\theta^\top \lambda \leq J \leq B$ at an optimum. Applying the grouped logical-constraint idea of [1] yields a shared fixed gate $\theta^\top w_j \leq B z_j$. Our problem-specific strengthening uses the risk being minimized: $J \leq \eta$ and $\theta^\top w_j \leq \eta z_j$. For fixed z this remains a convex oracle, and it preserves the exact binary objective.

The oracle's dual allocates source mass through Γ, measures unavoidable target overlap through $\mathcal{H}(\Gamma)$, and covers coordinatewise transport beyond class allocations with nonnegative prices μ_j. These expanded certificates need not be admissible original adversarial distributions. Scaling their mass and prices gives a normalization $\tau + \widehat{\mu}^\top z = 1$. The value function can then be written over a support-independent certificate domain as

$$R(z) = \max_c \frac{\mathcal{H}(\Gamma_c)}{1 + \mu_c^\top z}, \qquad R(z) = V(z) \text{ for binary } z. \tag{4}$$

Each fixed-certificate fraction is convex on $[0, 1]^D$. An oracle certificate generated at z^t therefore gives a global tangent. In normalized quantities, with its own value v_t, this is

$$\eta \geq L_t(z) := v_t - v_t(\widehat{\mu}^t)^\top (z - z^t). \tag{5}$$

A feasible nonoptimal certificate gives a valid, possibly non-touching cut. Its own recomputed dual value is essential; substituting a primal upper objective would invalidate that argument.

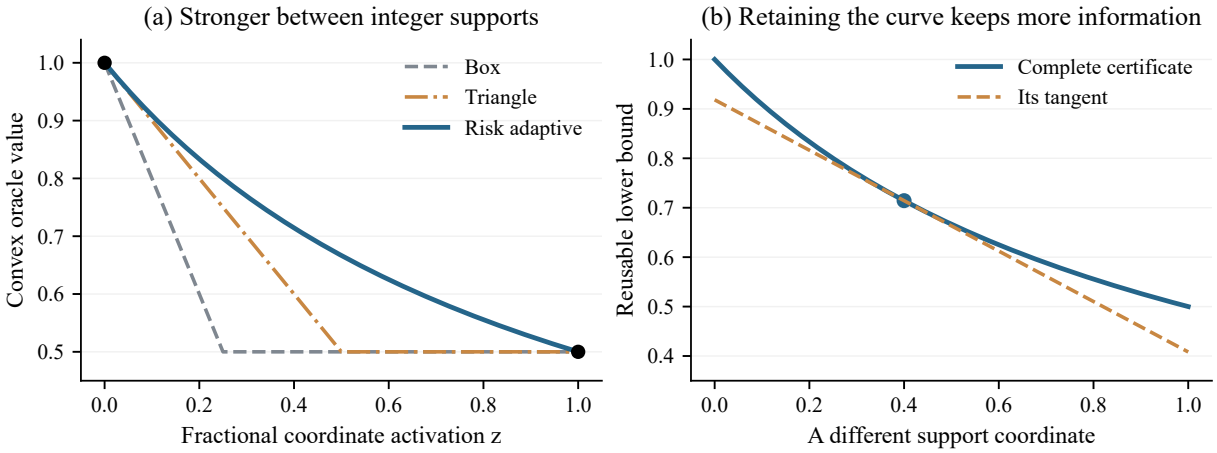


Figure 1: The gate changes the continuous lower model while preserving binary risk. The analytic one-coordinate example compares independent box, shared fixed-triangle, and risk-scaled gates. Values between zero and one describe relaxation weights, not partially observed physical measurements. The figure illustrates a theorem about oracle values; it does not predict runtime.

2.4 One complete oracle/master cycle

The master minimizes η over binary full-budget masks, $\eta \geq 0$, and all accumulated cuts (5). Its global solver bound is a lower bound L for (1). The best feasible original test has an upper bound U. Consequently,

$$L \leq \mathrm{OPT} \leq U, \qquad U - L \leq 10^{-6} + 10^{-5}|U| \tag{6}$$

is the declared numerical stopping criterion.

The loop is straightforward. Start from the same marginal-MMD screened support for every optimization variant. Refit every test variable and dual variable at that support. Keep its feasible-test upper bound and its global lower cut. Solve the master, obtain its next support, and repeat. Optional fractional root calls add lower cuts before integer search. They do not provide deployable-support upper bounds. Every new candidate is a complete continuous reoptimization; the method does not mix fixed historical transports as a substitute.

If the master is interrupted, its best feasible master objective is merely the value of a lower model at one support. The implementation uses the solver's *global bound*, not that incumbent objective, to update L. If a numerical oracle leaves a gap and the master revisits a cached support, the solver reports an accuracy limit. Exact finite termination is a theorem for tight oracle cuts and exact master solves; it is not inferred from repetition in floating-point computation.

3 Data are separated before any method is evaluated

3.1 Three disjoint roles within each training seed

Feature selection, downstream fitting, and final evaluation use separate samples. The selection split is the only split available to any selector. Its pooled mean and sample standard deviation define a common coordinate transformation, with a small scale floor for constant columns. That transformation is frozen and applied to every later split and every selector.

The downstream split fits two common heads on each selected support: the robust half-hinge envelope test and an RBF support-vector classifier. A selector cannot revise its support using this split. Final classification observations then assess those heads. Separate fresh two-sample batches assess equality testing; their labels

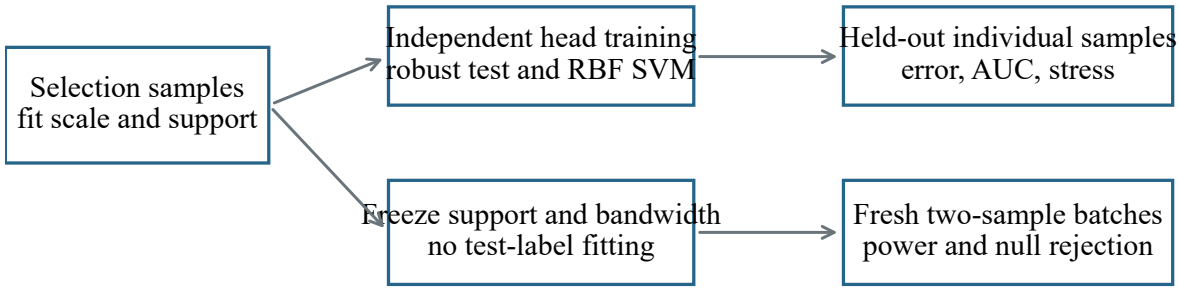


Figure 2: Data access in one synthetic training seed. Selection sees 20 observations per class; common heads see a disjoint 20 per class; classification uses 1,000 fresh observations per class. Twenty independent alternative batches and twenty null batches, each with 40 observations per class, assess a frozen support. This entire procedure is repeated over 30 training seeds. Conditional test batches are aggregated within a seed before uncertainty is computed across seeds.

Table 2: Synthetic mechanisms, before the selection-only standardization. Class zero is standard Gaussian. Each alternative changes only the three-coordinate signal block of class one.

Mechanism	Class-one signal block	What can reveal the change?
Mean	Add 0.9 to each signal coordinate.	Marginal means already contain information.
Variance	Multiply each signal coordinate by 1.8.	Marginal spread changes while means agree.
Dependence	Standard-normal marginals with equicorrelation 0.65.	Joint measurements are necessary; marginal population statistics agree.
Mixture	Add a common random sign times 1.5 to all three coordinates.	Marginal shape and joint mixture membership change; means agree.

are unavailable during selection and bandwidth fitting.

This separation makes the comparisons interpretable. Conditional on independently chosen measurements, the final permutation test has a fixed support and kernel. The common heads then assess every support through the same prediction pipeline, so differences in architecture cannot explain a selection comparison.

3.2 Four mechanisms ask four different questions

The statistical experiment has $D = 12$, three designated signal coordinates, and feature budget $d = 3$. All other coordinates are independent standard Gaussian noise. Feature positions are shuffled independently for each training seed. Recovery is evaluated in the underlying semantic coordinate indexing, so the shuffling does not artificially destroy stability.

The dependence case is particularly informative. Every individual coordinate has the same population distribution in both classes, so marginal MMD screening and an additive sum of marginal kernels have no population signal. Sampling fluctuation can nevertheless make their selected supports look informative in training. The held-out common RBF test can detect joint differences when the selected set contains useful signal coordinates. This separates the ability to choose a support from the ability of its downstream test to exploit interactions.

The mixture uses one shared sign per observation, not independently sampled signs for each coordinate. That distinction creates dependence among signal coordinates. It follows the role of equal-mean nonlinear alternatives in the source literature, without claiming a literal replay of the original high-dimensional experiment.

Thirty independent training seeds are predeclared. Each provides the three splits in Figure 2. The primary radii are $\theta_0 = \theta_1 = 0.25$ in the common standardized L1 geometry. Radii and scaling are not recomputed for individual supports. Otherwise two supports would be compared under different uncertainty sets, changing the discrete objective.

3.3 Two public real datasets

The UCI Wisconsin Diagnostic Breast Cancer dataset has 569 records and 30 numeric features; identifiers and diagnosis are excluded from the feature matrix. Class zero is benign and class one malignant. The UCI Ionosphere dataset has 351 records and 34 features; the label is excluded, with good returns mapped to zero and bad returns to one. The public source pages and persistent dataset identifiers are given in the references [8, 9].

For each of 20 stratified split seeds, 20 records per class are allocated to selection and another disjoint 20 per class to downstream fitting. Remaining records are held out for classification. The budget is three, the radii are 0.25, and robust global selection receives the same 30-second cap. Scaling is again fitted only on the selection records. Dataset downloads, label mappings, excluded fields, licensing, and file hashes are preserved in the source manifest.

Repeated splits reuse a finite dataset. Their mean and standard deviation describe split variability. Neither dataset has a known ground-truth feature support for this task, so synthetic recovery measures are omitted. These are methodological examples rather than clinical validation.

4 Each reported metric answers a stated question

4.1 Did the algorithm optimize its training objective?

The final upper bound is the objective of an independently feasible original test. The final lower bound comes from the global master solver. Their absolute difference measures unresolved optimization error; the normalized difference uses the reported objective scale. The numerical solved fraction counts runs meeting (6) out of all scheduled runs, including failures and time limits in the denominator.

Runtime measures total elapsed selection time, not only the last solver call. We also separate oracle time from master time, count all oracle calls including root points, and count distinct evaluated integer supports. A method may visit fewer supports while taking longer because each master search or oracle call is more expensive. Reporting these quantities together explains a runtime change instead of attributing it solely to a stronger relaxation.

The root bound measures what the finite set of fractional cuts has certified before integer search. The computational comparison separately examines the complete oracle value and this finite-cut master bound, since they measure different parts of the search.

4.2 Did the selected measurements support a calibrated equality test?

An equality test asks whether two new sample groups have the same distribution. This differs from labeling one observation. Given a frozen support, the common test computes the unbiased RBF maximum mean discrepancy statistic,

$$T = \frac{\sum_{i \neq i'} k(X_i, X_{i'})}{m(m-1)} + \frac{\sum_{h \neq h'} k(Y_h, Y_{h'})}{n(n-1)} - \frac{2 \sum_{i,h} k(X_i, Y_h)}{mn}. \tag{7}$$

The kernel uses all selected coordinates jointly. Its bandwidth is a median squared distance fitted on pooled selection data, then frozen. The final Gram matrix depends on the observations and frozen kernel, not on

how the test labels are shuffled.

Generate 199 random fixed-size label assignments. The p-value is

$$p = \frac{1 + \#\{T_{\text{perm}} \geq T_{\text{obs}}\}}{200}. \tag{8}$$

Reject at $p \leq 0.05$. Under the null, both fresh groups are drawn independently from class zero. Exchangeability conditional on the independently chosen support motivates this calibration. *Null rejection* is the empirical probability of rejection in these null batches; *power* is rejection in alternative batches. High power is convincing only alongside null rejection close to the nominal level. A small null excess should be interpreted with finite-repeat uncertainty, not automatically as a structural calibration failure.

Within each seed we average its twenty rejection indicators. Across-seed summaries and intervals use those seed-level fractions. The twenty conditional batches make each trained selector's assessment more precise; they do not create twenty independently trained selectors. The robust-risk theorem does not itself imply level-0.05 power. This common test evaluates the utility of the support, rather than claiming a new theorem connecting the two objectives.

4.3 Did the selector recover designated signal coordinates?

For selected set S and generating signal set T_*, report

$$\text{FDP} = \frac{|S \setminus T_*|}{\max(1, |S|)}, \qquad \text{NDP} = \frac{|T_* \setminus S|}{|T_*|}. \tag{9}$$

False-discovery proportion measures the fraction of purchased measurements outside the designated signal. Non-discovery proportion measures the signal fraction missed. Lower is better for both. Precision is $1 - \text{FDP}$, recall is $1 - \text{NDP}$, and F1 combines them as a harmonic mean. With equal selected and true support sizes, FDP and NDP coincide, while precision, recall and F1 all equal the recovered signal fraction. These equivalent views are retained for continuity with the source literature and do not supply independent findings.

Across-run stability is the mean pairwise Jaccard similarity $|S_a \cap S_b|/|S_a \cup S_b|$, using semantic coordinates in shuffled synthetic problems. Stability and recovery are different: a selector can consistently choose the same unhelpful variables. The designated changed-coordinate set is an interpretive reference, not a universal definition of a Bayes-optimal support in more general correlated data.

4.4 Did new individual observations get distinguished?

The common robust head gives a probability $\pi_0(x)$ of accepting class zero. On final observations, its randomized class-zero error is the mean of $1 - \pi_0$ and its randomized class-one error is the mean of π_0. Randomized balanced error is half their sum:

$$\text{BER}_{\text{rand}} = \frac{1}{2}\left[\frac{1}{m}\sum_i (1 - \pi_0(X_i)) + \frac{1}{n}\sum_h \pi_0(Y_h)\right]. \tag{10}$$

This equals the empirical half-hinge surrogate average. The surrogate sum is twice that value. These names describe the same measurement and its normalization, rather than independent successes.

For deterministic prediction, threshold π_0 at 0.5, assigning ties to class zero. Its classwise errors and balanced average are reported separately. The randomized certificate does not directly bound that thresholded error. Area under the ROC curve uses $1 - \pi_0$ as the class-one score and measures ranking across thresholds. Predictive log loss assesses probability sharpness; its probabilities are clipped only for that metric to avoid numerical infinities. A half-hinge-trained head has not thereby solved a logistic training objective.

The common RBF support-vector classifier supplies another balanced-error and ranking evaluation. It uses $C = 1$ and a median bandwidth fitted only on downstream-training data. Agreement across the two heads is more persuasive evidence about the support than a gain confined to the envelope construction.

4.5 What the uncertainty bars mean

Synthetic figures use 95% bootstrap intervals over independent training seeds, with 2,000 fixed-seed resamples. Test replicates are averaged inside their training seed first. Method differences are paired within seed because methods share data, transformations, test batches, and permutation seeds. Real-data summaries use mean $\pm$ standard deviation over repeated splits. The report does not treat a scan across many metrics and panels as a formal significance test, or reinterpret an optimization gap as sampling uncertainty.

5 Which baselines are reproduced, and what is held common?

All budget-matched selectors see exactly the same selection split and choose three coordinates in the primary statistical experiment. The subsequent heads, test data, preprocessing, and evaluation rules are common. Comparisons therefore target support utility. Each selector's criterion is explained below so that similarly named methods do not conceal different mathematical tasks.

5.1 Risk-based global selection and robust greedy

The proposed global method uses the exact two-class robust risk and global cuts from Section 2. A time-limited run contributes its best feasible support and actual bounds. Robust greedy shares the same loss and radius but adds the coordinate producing the smallest fully refitted robust risk at each forward stage. Its candidates are complete fixed-support solves. Greedy can miss a useful joint combination because an irreversible first choice constrains later stages. This baseline isolates the practical value of global search while holding the criterion fixed.

If greedy reaches its common wall limit before a full stage completes, it keeps its last completed support and fills the remaining budget with the predeclared marginal ranking. That outcome is labeled as a time-limited greedy adaptation. It is not silently recorded as an exact greedy trajectory.

5.2 Marginal screening, sparse logistic ranking, and random selection

Univariate screening ranks unbiased MMD values of one-dimensional RBF coordinate kernels. It is simple, transparent, and can perform strongly when marginal differences exist. It has no population signal in the equal-marginal dependence mechanism. It is not a reproduction of Wang et al.'s optimized weighted kernel.

Sparse logistic ranking fits the fixed L1 logistic baseline family, with liblinear and $C = 1$, on selection data. The largest absolute coefficients provide the common feature budget. Training variance and feature index resolve coefficient ties. Ranking a regularized fit to obtain exactly d coordinates is an explicit adaptation. A linear predictor is naturally challenged by some equal-mean mechanisms; this does not make its performance a universal statement about sparse classification.

Random selection chooses a seeded uniform budget-sized subset. It supplies a negative control under the same evaluation. On synthetic data, the known signal support and an all-feature test are additional diagnostics. The former is unavailable to a real selector; the latter buys more measurements. They are not budget-matched deployable competitors.

5.3 Wang additive-MMD: a criterion reproduction with disclosed differences

Wang, Dey, and Xie [4] optimize weights of a sparse additive kernel,

$$K_\omega(x,y) = \sum_j \omega_j \exp\left[-\frac{(x_j - y_j)^2}{2b_j^2}\right], \qquad \|\omega\|_2 = 1, \quad \|\omega\|_0 \le d. \tag{11}$$

"Linear" here means linear in kernel weights, not a raw dot-product kernel. Coordinate bandwidths follow the inspected revision convention: median squared cross-class selection distances, with factor one and a numerical floor for constant coordinates.

From paired coordinate-kernel contrasts, form the unbiased mean vector t and the row-covariance variance estimator $\widehat{\Sigma}$. The paper's mean-minus-variance criterion is

$$\max_{\|\omega\|_2=1,\ \|\omega\|_0\le d} t^\top \omega - 0.1\,\omega^\top \widehat{\Sigma}\omega. \tag{12}$$

For $D \le 12$, the implementation enumerates candidate supports and globally solves the sphere-constrained quadratic weight problem on each support. It includes the trust-region hard case rather than relying on a local gradient optimizer. For larger D, a documented forward/swap search optimizes the same criterion and is explicitly marked approximate.

The inspected official revision returns a *positive* variance matrix before an eigenvalue shift. The inspected optimizer maximizes the resulting positive quadratic term; it does not negate that matrix. The combination therefore rewards variance, whereas (12) penalizes it. Our primary baseline is labeled **Wang additive-MMD, paper objective**; it is not called a literal replay of the revision script. The official source commit and coefficient tests preserve the discrepancy for inspection [5].

The source testing routine also recomputes a cross-class bandwidth using final labels. Our native additive diagnostic freezes the training bandwidth and uses the common plus-one permutation construction, preserving the independent-test calibration logic. Optimized negative weights are retained where permitted by the criterion. Such a signed weighted statistic can be permutation-calibrated when frozen independently, but it should not be called a positive-definite-kernel MMD unless its weights are nonnegative.

The *common-support* test discards those weights and evaluates the selected coordinates with the same joint RBF statistic as every selector. The *native additive* diagnostic uses the learned weights. Their different labels distinguish support utility from a complete learned testing pipeline. Neither is presented as a reproduced numerical table from the source paper.

5.4 Why some nearby papers do not become automatic competitors

Swaroop et al.'s selector protects several specified downstream populations. Treating class zero and class one as those populations creates one-class groups and changes its problem. We adopt its evaluation separation without claiming to reproduce its group objective. Similarly, Hanada et al. [7] retain a safe union of potentially useful variables under adaptation; this need not satisfy a fixed acquisition budget. Neither method is silently transformed into a budget-matched comparator.

6 Correctness evidence before performance claims

The mathematical unit checks compare the original fixed-support primal with an independent source-marginal, total-transport-budget dual. This reference contains no coordinate gates, normalized mass scale, beta allocation, or mu prices. Agreement therefore tests the original objective rather than merely comparing two versions of the same matrix assembly.

Twenty random small instances use $D = 6$, three or four atoms per class, unequal positive radii, nonuniform positive weights, and budgets one through three. All supports are enumerated. Fixed gate primals, gate duals, original transport values, and every cross-support cut are compared. Random fractional points additionally check gate ordering and tangent validity. The suite includes duplicates, unequal class sizes, zero/full budgets, and analytic one-coordinate curves. Logistic uses smaller independently interpretable checks and an exponential-cone solver.

Deployment checks query the whole-space envelope, verify its probability interval, alter unselected coordinates, and move source atoms with an explicitly budget-feasible coupling. Additional experiment checks verify split independence, semantic recovery under shuffled positions, scalar kernel/variance coefficients, global weight fitting including the hard case, direct unbiased MMD arithmetic, plus-one p-values, and risk normalization. A simulated interrupted master confirms that its feasible incumbent is not substituted for its global lower bound.

These tests provide evidence about the actual implementation. The derivations establish the mathematical statements under their assumptions. Ordinary floating-point repair and small conservative margins are numerical safeguards, not a directed-rounding theorem for arbitrary ill-conditioned inputs. The report consequently uses declared numerical tolerances when describing optimality.

7 Computational results: stronger values, actual search costs

7.1 A paired comparison with the same objective

The primary comparison uses fixed box gates, shared fixed-triangle gates, risk-scaled gates without root cuts, and risk-scaled gates with five root rounds. Small cases also use enumeration. All compare the same integer risk, common initial support, radii, stopping rule, hardware, and 30-second limit. A solved gate variant should agree with independently completed enumeration, up to declared tolerance; a different objective is not a benefit of the gate.

The small design uses mean and mixture data at $D = 8, d = 2$ and $D = 12, d = 3$, eight selection atoms per class, and ten paired seeds. The scaling design uses $D = 20, 40, 80$, budget three, ten atoms per class, and five paired seeds. Each source distribution is standardized using its common selection pool. Total elapsed time includes root work, oracle construction/solves, every master search, and any numerical retry.

The next figures must be read together. Runtime without solved fraction can reward a method that fails early. Solved fraction without feasible objectives and remaining gaps can conceal how useful an unfinished run is. Distinct supports and the oracle/master decomposition explain where time is spent.

Mean difference — computational readout. At $D = 12$, Global risk solves 10/10 recorded instances, with mean charged time 16.54 s; Triangle solves 10/10 recorded instances, with mean charged time 4.63 s; Box solves 10/10 recorded instances, with mean charged time 4.07 s. Across the recorded $D \geq 20$ risk-method instances, 15/15 end at the time limit (100.0%). These time and solved-count comparisons retain unsuccessful outcomes; a stronger continuous gate has not removed the integer-search cost.

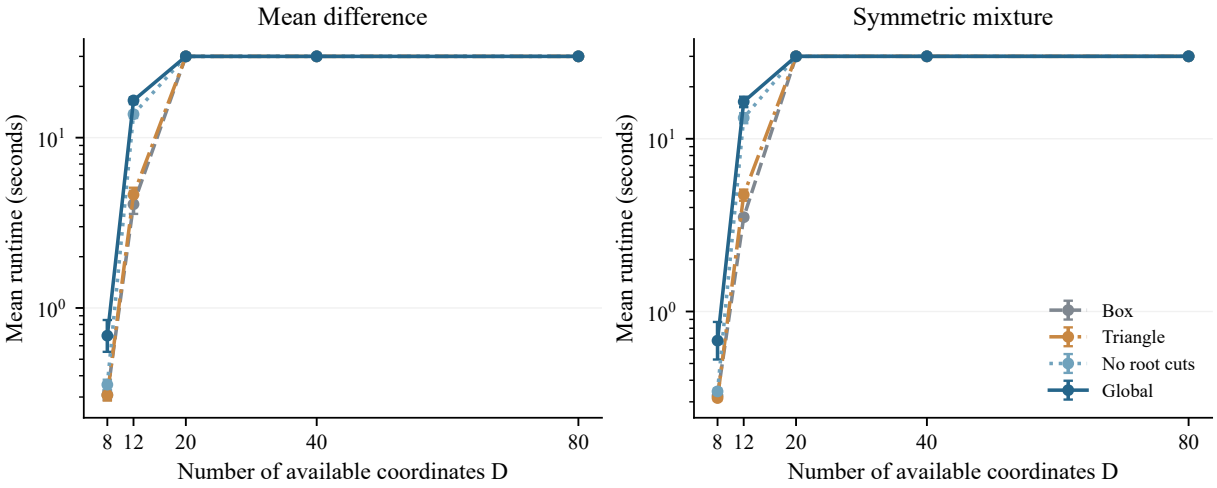


Figure 3: Primary computational comparison under a common 30-second selection budget. Total elapsed time includes unsuccessful runs, oracle work, and master work. Paired seeds share each instance and warm support. Synthetic uncertainty, where shown, is a 95% training-seed bootstrap interval; small and scaling designs have their separately predeclared seed counts. A log time axis separates subsecond solves from time-limited searches.

Table 3: Mean difference: primary gate comparison. Time is charged mean seconds [95% seed-bootstrap interval], with all outcomes retained. Gap is mean absolute upper-minus-lower bound; visits count distinct integer supports. O: numerical optimum; T: time limit; M: master failure; F: oracle failure; A: accuracy limit.

Setting / method	Solved / n	Time (s) ↓	Gap ↓	Visits	Status
$D = 8, d = 2$ Box	10/10	0.31 [0.29, 0.33]	4.0×10^{-7}	28.0	10O
$D = 8, d = 2$ Triangle	10/10	0.31 [0.29, 0.33]	6.0×10^{-7}	28.0	10O
$D = 8, d = 2$ Risk, no root	10/10	0.35 [0.33, 0.38]	4.0×10^{-7}	28.0	10O
$D = 8, d = 2$ Global risk	9/10	0.69 [0.55, 0.85]	0.011	27.3	1M, 9O
$D = 8, d = 2$ Enumeration	10/10	0.16 [0.16, 0.17]	1.4×10^{-9}	28.0	10O
$D = 12, d = 3$ Box	10/10	4.07 [3.57, 4.88]	1.0×10^{-7}	220.0	10O
$D = 12, d = 3$ Triangle	10/10	4.63 [4.16, 5.08]	1.3×10^{-9}	219.9	10O
$D = 12, d = 3$ Risk, no root	10/10	13.73 [13.05, 14.32]	4.0×10^{-7}	203.2	10O
$D = 12, d = 3$ Global risk	10/10	16.54 [15.74, 17.28]	2.0×10^{-7}	197.8	10O
$D = 12, d = 3$ Enumeration	10/10	1.38 [1.34, 1.40]	1.3×10^{-9}	220.0	10O
$D = 20, d = 3$ Box	0/5	30.01 [30.00, 30.02]	0.276	808.2	5T
$D = 20, d = 3$ Triangle	0/5	30.02 [30.01, 30.03]	0.279	812.2	5T
$D = 20, d = 3$ Risk, no root	0/5	30.00 [30.00, 30.00]	0.252	306.4	5T
$D = 20, d = 3$ Global risk	0/5	30.00 [30.00, 30.00]	0.167	178.8	5T
$D = 40, d = 3$ Box	0/5	30.02 [30.02, 30.03]	0.287	653.0	5T

Continued on next page

Continued

Setting / method	Solved / n	Time (s) ↓	Gap ↓	Visits	Status
$D = 40, d = 3$ Triangle	0/5	30.01 [30.01, 30.02]	0.287	693.6	5T
$D = 40, d = 3$ Risk, no root	0/5	30.04 [30.02, 30.06]	0.292	584.0	5T
$D = 40, d = 3$ Global risk	0/5	30.00 [30.00, 30.01]	0.199	122.8	5T
$D = 80, d = 3$ Box	0/5	30.01 [30.01, 30.02]	0.283	406.4	5T
$D = 80, d = 3$ Triangle	0/5	30.02 [30.01, 30.02]	0.299	447.4	5T
$D = 80, d = 3$ Risk, no root	0/5	30.05 [30.02, 30.07]	0.279	458.2	5T
$D = 80, d = 3$ Global risk	0/5	30.00 [30.00, 30.01]	0.179	90.2	5T

Symmetric mixture — computational readout. At $D = 12$, Global risk solves 10/10 recorded instances, with mean charged time 16.39 s; Triangle solves 10/10 recorded instances, with mean charged time 4.75 s; Box solves 10/10 recorded instances, with mean charged time 3.52 s. Across the recorded $D \geq 20$ risk-method instances, 15/15 end at the time limit (100.0%). These time and solved-count comparisons retain unsuccessful outcomes; a stronger continuous gate has not removed the integer-search cost.

Table 4: Symmetric mixture: primary gate comparison. Time is charged mean seconds [95% seed-bootstrap interval], with all outcomes retained. Gap is mean absolute upper-minus-lower bound; visits count distinct integer supports. O: numerical optimum; T: time limit; M: master failure; F: oracle failure; A: accuracy limit.

Setting / method	Solved / n	Time (s) ↓	Gap ↓	Visits	Status
$D = 8, d = 2$ Box	10/10	0.33 [0.31, 0.34]	1.0×10^{-7}	28.0	10O
$D = 8, d = 2$ Triangle	10/10	0.32 [0.31, 0.33]	4.0×10^{-7}	27.9	10O
$D = 8, d = 2$ Risk, no root	10/10	0.34 [0.33, 0.36]	6.0×10^{-7}	28.0	10O
$D = 8, d = 2$ Global risk	10/10	0.68 [0.53, 0.87]	1.3×10^{-9}	26.7	10O
$D = 8, d = 2$ Enumeration	10/10	0.17 [0.16, 0.17]	1.3×10^{-9}	28.0	10O
$D = 12, d = 3$ Box	10/10	3.52 [3.40, 3.63]	1.3×10^{-9}	220.0	10O
$D = 12, d = 3$ Triangle	10/10	4.75 [4.38, 5.08]	1.0×10^{-7}	219.2	10O
$D = 12, d = 3$ Risk, no root	10/10	13.22 [12.30, 14.07]	1.0×10^{-7}	200.0	10O
$D = 12, d = 3$ Global risk	10/10	16.39 [15.24, 17.52]	1.0×10^{-7}	197.0	10O
$D = 12, d = 3$ Enumeration	10/10	1.37 [1.35, 1.39]	1.3×10^{-9}	220.0	10O
$D = 20, d = 3$ Box	0/5	30.01 [30.00, 30.02]	0.263	804.0	5T
$D = 20, d = 3$ Triangle	0/5	30.03 [30.01, 30.05]	0.256	806.0	5T
$D = 20, d = 3$ Risk, no root	0/5	30.00 [30.00, 30.00]	0.236	301.8	5T

Continued on next page

Continued

Setting / method	Solved / n	Time (s) ↓	Gap ↓	Visits	Status
$D = 20, d = 3$		30.00			
Global risk	0/5	[30.00, 30.00]	0.151	188.8	5T
$D = 40, d = 3$		30.01			
Box	0/5	[30.00, 30.02]	0.263	655.8	5T
$D = 40, d = 3$		30.01			
Triangle	0/5	[30.00, 30.02]	0.283	692.6	5T
$D = 40, d = 3$		30.04			
Risk, no root	0/5	[30.02, 30.06]	0.268	582.2	5T
$D = 40, d = 3$		30.00			
Global risk	0/5	[30.00, 30.00]	0.176	127.2	5T
$D = 80, d = 3$		30.01			
Box	0/5	[30.00, 30.01]	0.287	419.2	5T
$D = 80, d = 3$		30.01			
Triangle	0/5	[30.00, 30.03]	0.288	460.4	5T
$D = 80, d = 3$		30.02			
Risk, no root	0/5	[30.01, 30.04]	0.291	442.2	5T
$D = 80, d = 3$		30.00			
Global risk	0/5	[30.00, 30.01]	0.190	96.8	5T

On 159 solved gate/enumeration pairs, the maximum absolute upper-objective difference is 0.000. Interval incompatibilities at combined numerical tolerances: 0. Enumeration references are paired by mechanism, dimension, budget, radius, and seed. A comparison with no completed reference is not counted as exact objective agreement.

How to read the evidence. The stronger risk-scaled extension is a valid theoretical result, while a runtime advantage is an empirical result. The primary records do not show a uniform runtime advantage. Enumeration can be cheaper in very small problems, and repeated master searches can dominate the global-cut implementation. A smaller number of visited supports is therefore not sufficient evidence of faster optimization.

7.2 Why the present master can be expensive

Each SciPy integer-master call starts a new mixed-integer search. The interface does not embed these cuts as lazy constraints in one continuing branch-and-bound tree. The reference paper discusses that implementation route, but it is not exposed by the present high-level interface. Restarting many master searches creates overhead even when the continuous oracle is inexpensive.

Moreover, the master retains tangents, not the entire function in (4). A tangent of a larger convex function can lie below a tangent of a smaller function away from the anchor. Theoretical pointwise dominance of complete extensions does not remove this limitation. The timing decomposition and number of nearly exhaustive support visits are the appropriate evidence for this explanation; they do not prove that all possible exact implementations would have the same bottleneck.

7.3 Numerical master failures and the documented remedy

Original runs exposed reproducible status-4 errors in the numerical master presolve. These outcomes must remain visible. The remedy retries the *identical* master model with presolve disabled, preserving the objective, constraints, cuts, and integer domain. The original attempt is retained in diagnostics and its time is charged to the same wall budget. This is a numerical implementation remedy rather than a new algorithmic treatment.

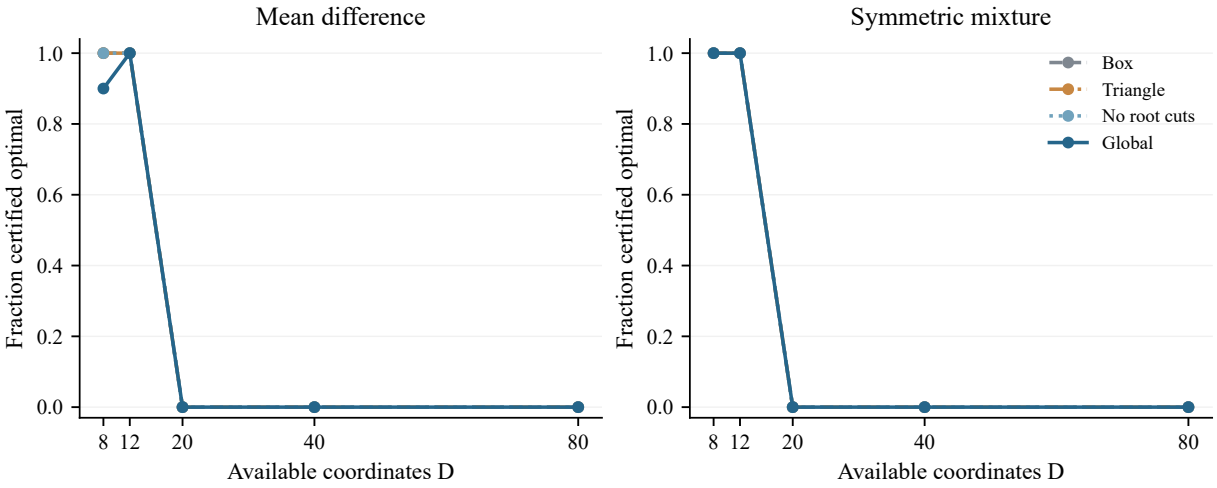


Figure 4: Numerical solved fraction for the same primary computational instances. "Solved" means the reported global lower and feasible upper bounds meet the declared tolerance. All scheduled outcomes belong in its denominator, including time limits and failures. Search effort is reported separately in the tables. A stronger oracle extension need not produce a faster finite-cut master.

The preserved original whole-run failure inventory contains 14 cases; 12 original records are archived. The current collection contains 12 corresponding rerun records, of which 11 no longer end in a master/oracle failure. There are 3 unresolved master/oracle failures. Final-run diagnostics record 66 identical-model master retries. A retry-resolved run can still end at a time or accuracy limit; resolution is not synonymous with numerical optimality.

Archived failed attempts charged 112.47 selection seconds and 112.87 end-to-end seconds. The final record timings include those original costs in addition to rerun costs. The rerun receives only the original 30-second selection allowance minus the original charged selection time; both attempts share that allowance, subject to modest call/finalization overhead. The appendix lists the retained original and final statuses rather than suppressing the first outcomes. The deterministic policy changes only presolve on a single retry of status 4, with the same model and remaining per-attempt budget. Status 4 supplies neither an incumbent nor a lower bound. The failed original call and the retry are charged to master and total selection time. The independent probe and policy are documented in the numerical remediation record; its one-off timing is not pooled into the experiment tables.

A completed retry has its own resulting bound, status, and charged time. Summaries identify whether they use original attempts or remedied outcomes; time-limited and accuracy-limited results retain their feasible tests and unresolved gaps.

8 Statistical results: calibration, power, recovery, and prediction

8.1 Power must be read alongside null rejection

The equality-test panel asks whether the selected coordinates reveal a population difference in fresh batches. The null panel asks whether the same frozen-support procedure rejects too often when both groups come from class zero. All selectors share test batches and permutation seeds within a training seed. This pairing prevents a difference in sampled difficulty from being attributed to selection.

The mean and variance mechanisms give marginal selectors a real population signal. Dependence gives them no such signal, making it a useful countercheck against overgeneralizing their strengths. The mixture has both marginal shape information and joint structure. We therefore interpret each mechanism rather than

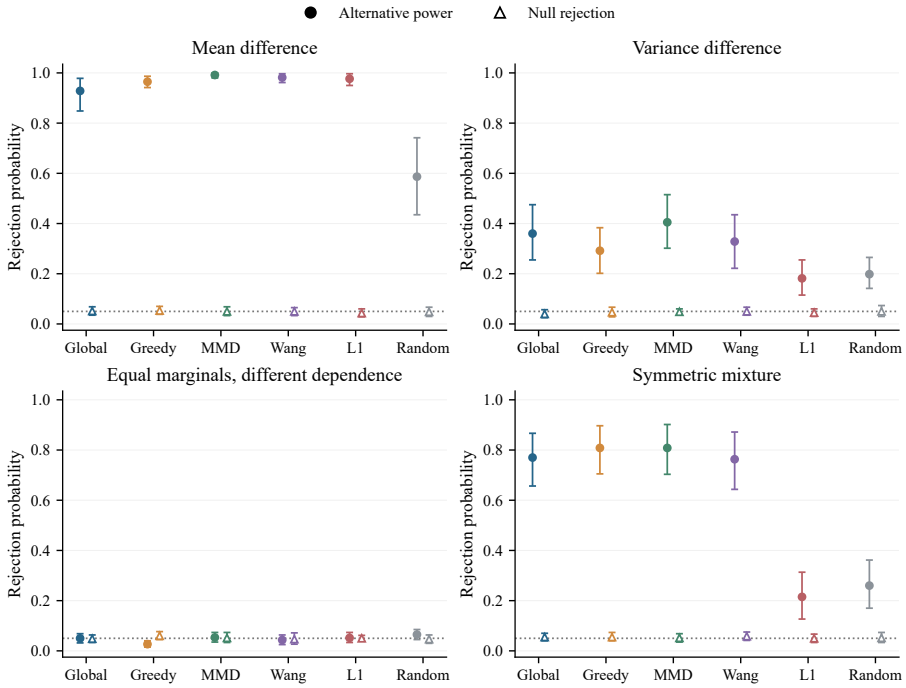


Figure 5: Independent two-sample testing on frozen selected supports. Alternative power and null rejection are reported together at nominal level 0.05, with 199 random permutations and plus-one p-values. Each point first averages twenty fresh batches within a training seed; 95% bootstrap intervals resample training seeds from the thirty-seed design, with effective counts in the tables. Diagnostic all-feature and known-signal references are distinct from budget-matched selectors.

compressing all four into an unexplained average rank.

The primary mean, variance, and dependence comparisons do not show an advantage for the robust-risk selector. Other selectors can obtain better measured power or recovery. The training objective minimizes a radius-dependent robust classification surrogate, whereas the final statistic measures a calibrated distribution difference. These findings make the objective's finite-sample usefulness an open issue even when its empirical minimization is accurate.

8.2 Signal recovery explains part of the power differences

If recovery and power both worsen, missing changed coordinates is a plausible explanation. If recovery is similar but power differs, the sampled training rule, the selected joint structure, or the downstream bandwidth may matter. The known-signal diagnostic shows what the common test can do on the designated support under the same finite sample sizes. It is an informative reference, rather than a proven upper bound on power or a support available to a deployable selector.

Mean difference — what the selected coordinates reveal. Observed common-test power is 0.928 for global robust-risk selection (30 valid training seeds) and 0.992 for marginal MMD screening (30 seeds). Their within-seed global-minus-screen difference is -6.3 percentage points, with a 95% seed-bootstrap interval [-14.3, -1.3] over 30 paired seeds. The mean fraction of the designated signal recovered is 0.533 for global selection (30 seeds) versus 0.756 for screening (30 seeds). Power is a batch-detection probability; recovered fraction describes which coordinates were chosen. Reading them together connects detection performance to support recovery without asserting causation or a formal significance finding. Global selection meets the optimization tolerance in 30/30 runs, yet its common-test power is 0.928 versus 0.965 for robust greedy. Empirical robust-risk optimality targets the training criterion; it need not maximize indepen-

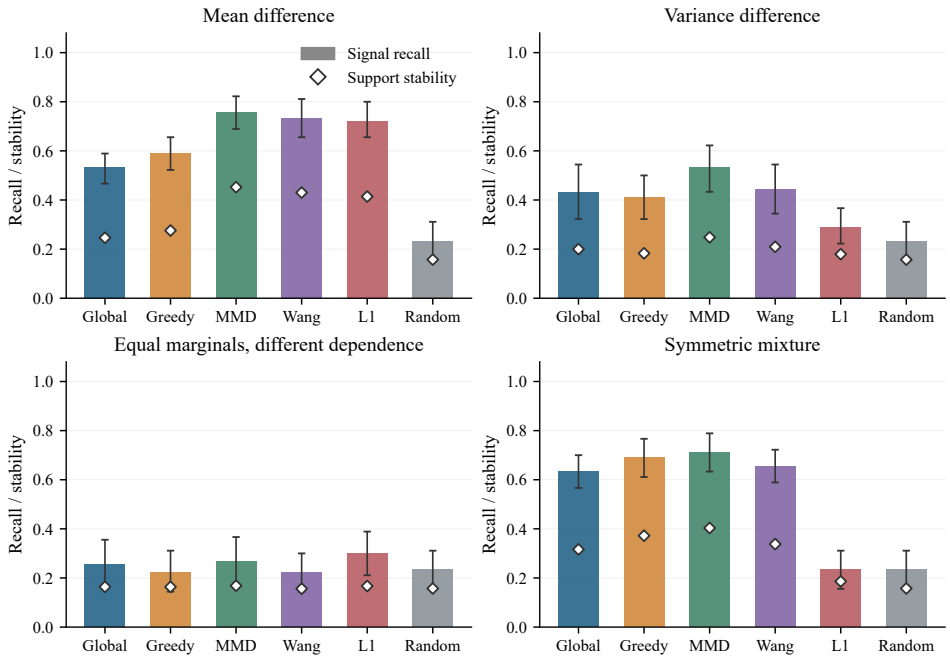


Figure 6: Recovery of the three designated signal coordinates under budget three. Bars show signal recall (higher is better), with 95% training-seed bootstrap intervals. Diamonds show descriptive mean pairwise support-stability Jaccard, without uncertainty bars. Positions are shuffled for selection but mapped back to semantic coordinates for recovery and stability. FDP and NDP are reported in the tables; equal true and selected support sizes make both equal to one minus recall.

dent testing power. For Wang's paper-objective baseline, the permutation-calibrated native signed additive statistic has observed power 0.977 and null rejection 0.045 (30 valid training seeds). Its support-only common RBF test has power 0.982. These evaluate different testing procedures; common-support power is not native Wang power.

Table 5: Mean difference: common-support testing and signal recovery. Values are means [95% training-seed bootstrap intervals]. The nominal null rejection level is 0.05. FDP and NDP coincide at this equal budget and signal size.

Selector	n	Power ↑	Null rejection	FDP ↓	F1 ↑
Global risk	30	0.928 [0.848, 0.978]	0.052 [0.035, 0.068]	0.467 [0.411, 0.533]	0.533 [0.467, 0.589]
Robust greedy	30	0.965 [0.942, 0.987]	0.053 [0.038, 0.070]	0.411 [0.344, 0.478]	0.589 [0.522, 0.656]
MMD screening	30	0.992 [0.980, 1.000]	0.050 [0.033, 0.068]	0.244 [0.178, 0.311]	0.756 [0.689, 0.822]
Additive MMD	30	0.982 [0.962, 0.997]	0.050 [0.033, 0.065]	0.267 [0.189, 0.344]	0.733 [0.656, 0.811]
L1 logistic	30	0.977 [0.950, 0.997]	0.043 [0.028, 0.060]	0.278 [0.200, 0.344]	0.722 [0.656, 0.800]
Random	30	0.587 [0.435, 0.742]	0.048 [0.030, 0.067]	0.767 [0.689, 0.844]	0.233 [0.156, 0.311]

Table 6: Mean difference: independently refitted common heads. Randomized and thresholded balanced error are distinct; AUC evaluates ranking. Values are means [95% seed intervals].

Selector	n	Random BER ↓	BER ↓	AUC ↑	SVM BER ↓
Global risk	30	0.448 [0.437, 0.459]	0.382 [0.360, 0.404]	0.650 [0.625, 0.676]	0.315 [0.297, 0.336]
Robust greedy	30	0.447 [0.437, 0.457]	0.380 [0.359, 0.400]	0.657 [0.633, 0.681]	0.306 [0.288, 0.325]
MMD screening	30	0.433 [0.425, 0.442]	0.346 [0.329, 0.364]	0.694 [0.673, 0.714]	0.279 [0.265, 0.293]
Additive MMD	30	0.439 [0.429, 0.449]	0.360 [0.339, 0.383]	0.679 [0.654, 0.703]	0.286 [0.269, 0.306]
L1 logistic	30	0.436 [0.427, 0.446]	0.354 [0.334, 0.373]	0.688 [0.665, 0.710]	0.281 [0.266, 0.295]
Random	30	0.482 [0.474, 0.490]	0.456 [0.437, 0.474]	0.559 [0.536, 0.583]	0.405 [0.377, 0.435]

Variance difference — what the selected coordinates reveal. Observed common-test power is 0.360 for global robust-risk selection (30 valid training seeds) and 0.405 for marginal MMD screening (30 seeds). Their within-seed global-minus-screen difference is -4.5 percentage points, with a 95% seed-bootstrap interval [-19.7, +11.0] over 30 paired seeds. The mean fraction of the designated signal recovered is 0.433 for global selection (30 seeds) versus 0.533 for screening (30 seeds). Power is a batch-detection probability; recovered fraction describes which coordinates were chosen. Reading them together connects detection performance to support recovery without asserting causation or a formal significance finding. Global selection meets the optimization tolerance in 27/30 runs, yet its common-test power is 0.360 versus 0.292 for robust greedy. Empirical robust-risk optimality targets the training criterion; it need not maximize independent testing power. For Wang's paper-objective baseline, the permutation-calibrated native signed additive statistic has observed power 0.518 and null rejection 0.052 (30 valid training seeds). Its support-only common RBF test has power 0.328. These evaluate different testing procedures; common-support power is not native Wang power.

Table 7: Variance difference: common-support testing and signal recovery. Values are means [95% training-seed bootstrap intervals]. The nominal null rejection level is 0.05. FDP and NDP coincide at this equal budget and signal size.

Selector	n	Power ↑	Null rejection	FDP ↓	F1 ↑
Global risk	30	0.360 [0.255, 0.475]	0.040 [0.025, 0.057]	0.567 [0.456, 0.678]	0.433 [0.322, 0.544]
Robust greedy	30	0.292 [0.202, 0.383]	0.047 [0.028, 0.067]	0.589 [0.500, 0.678]	0.411 [0.322, 0.500]
MMD screening	30	0.405 [0.302, 0.515]	0.048 [0.037, 0.060]	0.467 [0.378, 0.567]	0.533 [0.433, 0.622]
Additive MMD	30	0.328 [0.222, 0.435]	0.050 [0.035, 0.067]	0.556 [0.456, 0.656]	0.444 [0.344, 0.544]
L1 logistic	30	0.182 [0.115, 0.255]	0.045 [0.030, 0.060]	0.711 [0.633, 0.778]	0.289 [0.222, 0.367]
Random	30	0.198 [0.142, 0.265]	0.050 [0.030, 0.073]	0.767 [0.689, 0.844]	0.233 [0.156, 0.311]

Table 8: Variance difference: independently refitted common heads. Randomized and thresholded balanced error are distinct; AUC evaluates ranking. Values are means [95% seed intervals].

Selector	n	Random BER ↓	BER ↓	AUC ↑	SVM BER ↓
Global risk	30	0.482 [0.476, 0.488]	0.486 [0.477, 0.496]	0.565 [0.543, 0.585]	0.420 [0.398, 0.443]
Robust greedy	30	0.479 [0.473, 0.484]	0.476 [0.467, 0.485]	0.574 [0.555, 0.594]	0.425 [0.404, 0.444]
MMD screening	30	0.472 [0.465, 0.478]	0.466 [0.454, 0.478]	0.595 [0.574, 0.616]	0.400 [0.378, 0.420]
Additive MMD	30	0.477 [0.470, 0.484]	0.476 [0.464, 0.489]	0.576 [0.554, 0.599]	0.418 [0.394, 0.442]
L1 logistic	30	0.484 [0.478, 0.489]	0.477 [0.468, 0.487]	0.555 [0.537, 0.573]	0.447 [0.430, 0.463]
Random	30	0.489 [0.484, 0.493]	0.487 [0.476, 0.496]	0.540 [0.525, 0.556]	0.454 [0.438, 0.470]

Equal marginals, different dependence — what the selected coordinates reveal. Observed common-test power is 0.050 for global robust-risk selection (30 valid training seeds) and 0.053 for marginal MMD screening (30 seeds). Their within-seed global-minus-screen difference is -0.3 percentage points, with a 95% seed-bootstrap interval [-2.8, +2.0] over 30 paired seeds. The mean fraction of the designated signal recovered is 0.256 for global selection (30 seeds) versus 0.267 for screening (30 seeds). Power is a batch-detection probability; recovered fraction describes which coordinates were chosen. Reading them together connects detection performance to support recovery without asserting causation or a formal significance finding. Global selection meets the optimization tolerance in 30/30 runs, yet its common-test power is 0.050 versus 0.027 for robust greedy. Empirical robust-risk optimality targets the training criterion; it need not maximize independent testing power. For Wang's paper-objective baseline, the permutation-calibrated native signed additive statistic has observed power 0.045 and null rejection 0.052 (30 valid training seeds). Its support-only common RBF test has power 0.043. These evaluate different testing procedures; common-support power is not native Wang power.

Table 9: Equal marginals, different dependence: common-support testing and signal recovery. Values are means [95% training-seed bootstrap intervals]. The nominal null rejection level is 0.05. FDP and NDP coincide at this equal budget and signal size.

Selector	n	Power ↑	Null rejection	FDP ↓	F1 ↑
Global risk	30	0.050 [0.032, 0.068]	0.048 [0.033, 0.063]	0.744 [0.644, 0.844]	0.256 [0.156, 0.356]
Robust greedy	30	0.027 [0.015, 0.040]	0.060 [0.045, 0.077]	0.778 [0.689, 0.856]	0.222 [0.144, 0.311]
MMD screening	30	0.053 [0.035, 0.073]	0.052 [0.033, 0.073]	0.733 [0.633, 0.822]	0.267 [0.178, 0.367]
Additive MMD	30	0.043 [0.025, 0.063]	0.047 [0.027, 0.072]	0.778 [0.700, 0.856]	0.222 [0.144, 0.300]
L1 logistic	30	0.052 [0.033, 0.073]	0.050 [0.038, 0.062]	0.700 [0.611, 0.789]	0.300 [0.211, 0.389]
Random	30	0.065 [0.045, 0.085]	0.047 [0.030, 0.063]	0.767 [0.689, 0.844]	0.233 [0.156, 0.311]

Table 10: Equal marginals, different dependence: independently refitted common heads. Randomized and thresholded balanced error are distinct; AUC evaluates ranking. Values are means [95% seed intervals].

Selector	n	Random BER ↓	BER ↓	AUC ↑	SVM BER ↓
Global risk	30	0.497 [0.494, 0.500]	0.492 [0.483, 0.499]	0.510 [0.500, 0.520]	0.494 [0.488, 0.500]
Robust greedy	30	0.499 [0.497, 0.501]	0.496 [0.490, 0.503]	0.504 [0.497, 0.512]	0.492 [0.486, 0.498]
MMD screening	30	0.497 [0.494, 0.500]	0.491 [0.481, 0.499]	0.509 [0.501, 0.521]	0.490 [0.482, 0.497]
Additive MMD	30	0.497 [0.495, 0.499]	0.494 [0.488, 0.500]	0.509 [0.502, 0.516]	0.493 [0.486, 0.500]
L1 logistic	30	0.498 [0.496, 0.500]	0.494 [0.486, 0.500]	0.507 [0.499, 0.515]	0.494 [0.489, 0.499]
Random	30	0.499 [0.497, 0.501]	0.496 [0.490, 0.502]	0.503 [0.497, 0.509]	0.497 [0.492, 0.502]

Symmetric mixture — what the selected coordinates reveal. Observed common-test power is 0.770 for global robust-risk selection (30 valid training seeds) and 0.808 for marginal MMD screening (30 seeds). Their within-seed global-minus-screen difference is -3.8 percentage points, with a 95% seed-bootstrap interval [-20.3, +11.7] over 30 paired seeds. The mean fraction of the designated signal recovered is 0.633 for global selection (30 seeds) versus 0.711 for screening (30 seeds). Power is a batch-detection probability; recovered fraction describes which coordinates were chosen. Reading them together connects detection performance to support recovery without asserting causation or a formal significance finding. Global selection meets the optimization tolerance in 27/30 runs, yet its common-test power is 0.770 versus 0.808 for robust greedy. Empirical robust-risk optimality targets the training criterion; it need not maximize independent testing power. For Wang's paper-objective baseline, the permutation-calibrated native signed additive statistic has observed power 0.945 and null rejection 0.040 (30 valid training seeds). Its support-only common RBF test has power 0.763. These evaluate different testing procedures; common-support power is not native Wang power.

Table 11: Symmetric mixture: common-support testing and signal recovery. Values are means [95% training-seed bootstrap intervals]. The nominal null rejection level is 0.05. FDP and NDP coincide at this equal budget and signal size.

Selector	n	Power ↑	Null rejection	FDP ↓	F1 ↑
Global risk	30	0.770 [0.657, 0.867]	0.055 [0.040, 0.070]	0.367 [0.300, 0.433]	0.633 [0.567, 0.700]
Robust greedy	30	0.808 [0.705, 0.897]	0.055 [0.040, 0.073]	0.311 [0.233, 0.389]	0.689 [0.611, 0.767]
MMD screening	30	0.808 [0.703, 0.902]	0.052 [0.035, 0.068]	0.289 [0.211, 0.367]	0.711 [0.633, 0.789]
Additive MMD	30	0.763 [0.643, 0.872]	0.058 [0.042, 0.075]	0.344 [0.278, 0.411]	0.656 [0.589, 0.722]
L1 logistic	30	0.215 [0.127, 0.313]	0.050 [0.033, 0.067]	0.767 [0.689, 0.844]	0.233 [0.156, 0.311]
Random	30	0.260 [0.170, 0.362]	0.052 [0.033, 0.073]	0.767 [0.689, 0.844]	0.233 [0.156, 0.311]

Table 12: Symmetric mixture: independently refitted common heads. Randomized and thresholded balanced error are distinct; AUC evaluates ranking. Values are means [95% seed intervals].

Selector	n	Random BER ↓	BER ↓	AUC ↑	SVM BER ↓
Global risk	30	0.423 [0.408, 0.438]	0.350 [0.322, 0.381]	0.721 [0.686, 0.753]	0.314 [0.290, 0.340]
Robust greedy	30	0.411 [0.395, 0.427]	0.325 [0.294, 0.358]	0.746 [0.709, 0.780]	0.297 [0.269, 0.325]
MMD screening	30	0.405 [0.390, 0.420]	0.313 [0.285, 0.342]	0.754 [0.723, 0.785]	0.289 [0.262, 0.316]
Additive MMD	30	0.417 [0.403, 0.429]	0.332 [0.304, 0.358]	0.737 [0.707, 0.766]	0.314 [0.285, 0.342]
L1 logistic	30	0.479 [0.468, 0.488]	0.462 [0.438, 0.481]	0.568 [0.542, 0.599]	0.445 [0.424, 0.466]
Random	30	0.478 [0.467, 0.488]	0.458 [0.435, 0.477]	0.569 [0.542, 0.600]	0.439 [0.414, 0.462]

Table 13: Paired global-risk-minus-marginal-screening power. Negative values mean lower observed power for global robust-risk selection; the interval resamples training-seed differences.

Mechanism	Paired n	Δ power [95% CI]
Mean difference	30	-0.063 [-0.143, -0.013]
Variance difference	30	-0.045 [-0.197, 0.110]
Equal marginals, different dependence	30	-0.003 [-0.028, 0.020]
Symmetric mixture	30	-0.038 [-0.203, 0.117]

What the empirical head certificate covers. Across 120 valid synthetic global-selection training seeds, 120 independent randomized balanced errors exceed their own refitted head's empirical risk bound divided by two. Among the corresponding 120 valid shifted-source checks, 0 surrogate violations exceed 10^{-7}. The comparison below uses the independent head-training certificate, not the support-selection objective.

Table 14: Global selector: means of refitted empirical head bounds and independent randomized error, paired within training seed. The bound is divided by two to match balanced-average risk units. Above-bound counts use a 10^{-8} comparison tolerance. Source violation is the largest recorded direct budget-feasible shifted-source violation, rather than a population stress error.

Mechanism	Paired n	Head bound / 2	Held-out BER	Above bound	Source violation
Mean difference	30	0.251	0.448	30/30	0.000
Variance difference	30	0.275	0.482	30/30	0.000
Equal marginals, different dependence	30	0.287	0.497	30/30	0.000
Symmetric mixture	30	0.248	0.423	30/30	0.000

The fitted certificate protects distributions in its empirical Wasserstein balls. The evaluated populations were not shown to belong to those balls, so the held-out excess is not a certificate failure. Population coverage at the chosen sample size and radius remains unestablished. The head-training bound also differs from the selection objective because the head is refitted on a separate training split.

All six selectors above buy the same number of coordinates. Known-signal and all-feature diagnostics are reserved for the appendix. Within-seed two-sample batches were averaged before uncertainty was computed;

conditional batches are not counted as independent selections. Valid predictive metrics from a time-limited or failed global search describe its retained support and do not certify optimal selection.

8.3 Common-head classification supplies a different check

The classification table compares randomized envelope error, thresholded error, ranking, and the common RBF classifier. Each head is freshly fitted after selection on the same independent training split.

Thresholded error and randomized error can differ noticeably. A probability near 0.5 incurs nearly one-half randomized error but may be assigned to one class consistently after thresholding. Predictive log loss adds evidence about probability sharpness.

An apparent advantage confined to the robust head could reflect compatibility with that deployment rule. The common RBF head helps reveal that possibility. Conversely, lower classification error need not imply greater equality-test power, because classifying one observation and detecting a distribution difference from a batch have different statistical targets.

9 Stress and radius sensitivity: what robustness is actually tested?

9.1 Two different shifted-data calculations

The population stress experiment moves held-out observations toward the opposite class. A single common direction is fitted from the downstream-training means and normalized to L1 length one. For $\rho \in \{0, 0.25, 0.50\}$, class zero is translated by $+\rho v$ and class one by $-\rho v$. Every selector receives the same transformed observations and direction.

This experiment measures response to three predeclared population perturbations. Its largest measured error is a maximum over this finite set, not the exact Wasserstein supremum. A translation admits a coupling of cost ρ relative to its unshifted population; it does not establish inclusion in a ball around a finite empirical training distribution.

The direct check instead translates the downstream-training *source atoms* by at most their radius. The explicit source-to-shifted-source coupling proves membership in the empirical Wasserstein ball. Its surrogate sum can therefore be compared with the fitted primal risk bound. A positive numerical violation is a certificate issue to inspect; a population stress error above a training bound does not establish such an issue without population-ball inclusion.

9.2 The radius is a modeling choice, not a result-selected setting

The primary radius is fixed at 0.25 before the comparison. A separate predeclared sensitivity design uses radii $0.10, 0.25, 0.50$ for mean and mixture mechanisms on ten specified seeds. Supports are refitted at each radius, using the same seed-level datasets and common evaluation. No radius is chosen by looking at final test outcomes.

Mean difference — radius readout. From radius 0.10 to 0.50, observed power changes from 0.790 to 0.955, randomized balanced error from 0.494 to 0.426, and worst measured thresholded stress error from 0.488 to 0.380. The observed power averages increase across these settings. Both the selected support and the independently trained robust head are refitted at each radius. The sweep therefore describes their joint sensitivity; it does not isolate a causal effect of selection alone.

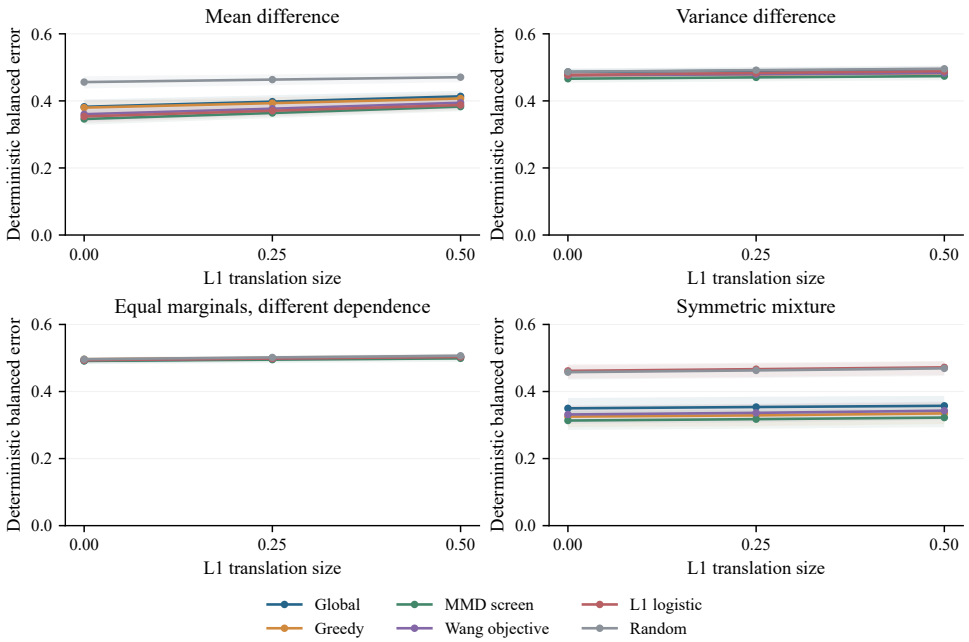


Figure 7: Held-out prediction under the common class-toward-class translations of L1 magnitude $\rho = 0, 0.25, 0.50$. Curves summarize paired stress observations, with 95% training-seed bootstrap intervals for synthetic data where shown. The maximum is over these specified shifts. No claim is made that the shifted populations belong to the empirical training Wasserstein balls. The separate shifted-source calculation is the direct certificate check.

Table 15: Mean difference: radius sensitivity, means [95% training-seed intervals]. The primary radius remains 0.25. Supports are refitted at each radius; no result-selected setting is declared optimal.

Radius	n	Power ↑	Random BER ↓	Stress BER ↓	Status
0.10	10	0.790 [0.595, 0.915]	0.494 [0.491, 0.497]	0.488 [0.478, 0.498]	10O
0.25	10	0.855 [0.650, 0.980]	0.466 [0.450, 0.480]	0.441 [0.409, 0.468]	10O
0.50	10	0.955 [0.915, 0.990]	0.426 [0.411, 0.440]	0.380 [0.354, 0.404]	10O

Symmetric mixture — radius readout. From radius 0.10 to 0.50, observed power changes from 0.495 to 0.970, randomized balanced error from 0.486 to 0.371, and worst measured thresholded stress error from 0.499 to 0.256. The observed power averages increase across these settings. Both the selected support and the independently trained robust head are refitted at each radius. The sweep therefore describes their joint sensitivity; it does not isolate a causal effect of selection alone.

Table 16: Symmetric mixture: radius sensitivity, means [95% training-seed intervals]. The primary radius remains 0.25. Supports are refitted at each radius; no result-selected setting is declared optimal.

Radius	n	Power ↑	Random BER ↓	Stress BER ↓	Status
0.10	10	0.495 [0.295, 0.710]	0.486 [0.478, 0.493]	0.499 [0.483, 0.514]	10O
0.25	10	0.630 [0.425, 0.845]	0.446 [0.417, 0.469]	0.402 [0.343, 0.450]	1M, 9O
0.50	10	0.970 [0.930, 0.995]	0.371 [0.352, 0.392]	0.256 [0.215, 0.300]	1M, 9O

Increasing a radius gives the adversary more freedom in the empirical optimization problem. It can change

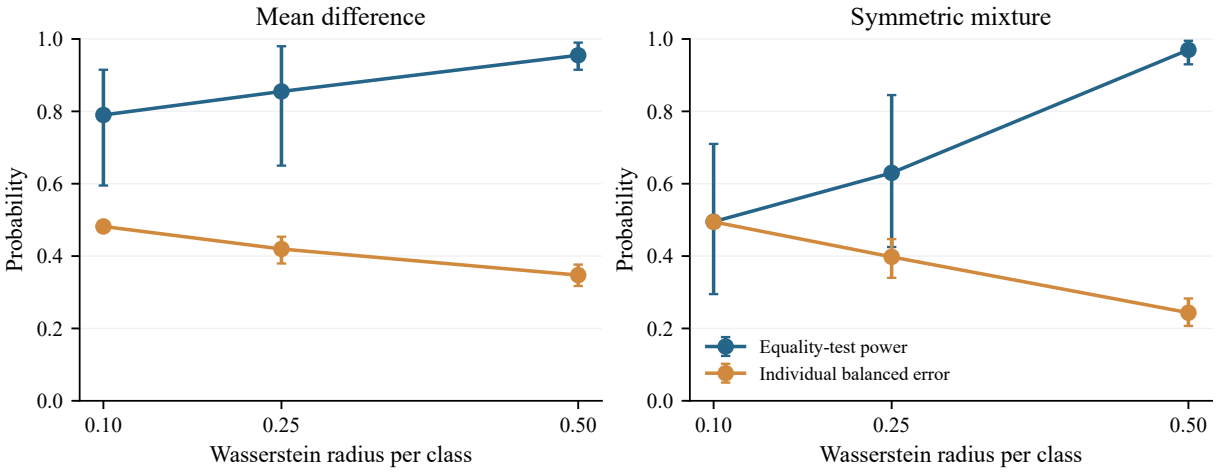


Figure 8: Sensitivity of risk-based selection to the predeclared radii $0.10, 0.25, 0.50$ on mean and mixture data. The primary setting remains 0.25. Each radius refits the support on the same seed-level selection data and is evaluated on the same independent observations; uncertainty resamples the ten training seeds. The sweep describes sensitivity and does not certify an optimal radius or population coverage.

transport prices and which support best balances overlap and distance. It does not necessarily improve finite-sample held-out error or measured stress performance. The sensitivity figure should therefore be read as a description of how the selected rule responds to a modeling assumption, not as a monotone robustness theorem about unknown populations.

10 Real-data results and the stability of selected measurements

Real-data evaluation keeps the same support budget and common prediction heads. The next table reports classification and finite-stress behavior together with selection time, bounds, and status. Results of a time-limited global search describe its incumbent support; they are not silently described as a globally best empirical subset.

Table 17: Wisconsin diagnostic breast cancer: mean ± split SD, not an independent-population confidence interval. Worst stress is the maximum over the three prescribed shifts. Effective metric counts and selection statuses are retained in the appendix.

Selector	n	Random BER ↓	BER ↓	AUC ↑	SVM BER ↓	Stress BER ↓
Global risk	20	0.251±0.046	0.112±0.031	0.965±0.020	0.081±0.017	0.119±0.030
Robust greedy	20	0.220±0.032	0.099±0.021	0.973±0.013	0.080±0.021	0.106±0.020
MMD screening	20	0.179±0.013	0.091±0.021	0.973±0.011	0.084±0.017	0.098±0.021
Additive MMD	20	0.180±0.013	0.090±0.019	0.975±0.010	0.082±0.014	0.096±0.019
L1 logistic	20	0.230±0.034	0.097±0.035	0.971±0.017	0.078±0.020	0.106±0.035
Random	20	0.310±0.066	0.192±0.089	0.893±0.093	0.156±0.078	0.199±0.088

Table 18: Wisconsin diagnostic breast cancer: charged selection time and original-risk bounds. A dash means no global original-objective certificate, not zero risk. Risk statuses use O/T/M/F/A as in the computational tables.

Selector	n	Time (s) ↓	Mean LB	Mean UB	Status
Global risk	20	30.00±0.00	0.143	0.296	20T
Robust greedy	20	0.54±0.02	—	—	baseline
MMD screening	20	0.00±0.00	—	—	baseline
Additive MMD	20	0.02±0.00	—	—	baseline
L1 logistic	20	0.00±0.00	—	—	baseline
Random	20	0.00±0.00	—	—	baseline

Table 19: Ionosphere: mean ± split SD, not an independent-population confidence interval. Worst stress is the maximum over the three prescribed shifts. Effective metric counts and selection statuses are retained in the appendix.

Selector	n	Random BER ↓	BER ↓	AUC ↑	SVM BER ↓	Stress BER ↓
Global risk	20	0.300±0.045	0.243±0.059	0.869±0.053	0.251±0.094	0.247±0.062
Robust greedy	20	0.285±0.035	0.217±0.052	0.885±0.045	0.202±0.081	0.223±0.052
MMD screening	20	0.284±0.044	0.223±0.055	0.865±0.055	0.195±0.053	0.229±0.053
Additive MMD	20	0.273±0.042	0.209±0.047	0.877±0.051	0.181±0.058	0.216±0.047
L1 logistic	20	0.284±0.034	0.227±0.050	0.880±0.046	0.198±0.049	0.232±0.047
Random	20	0.330±0.028	0.270±0.047	0.835±0.034	0.303±0.069	0.276±0.048

Table 20: Ionosphere: charged selection time and original-risk bounds. A dash means no global original-objective certificate, not zero risk. Risk statuses use O/T/M/F/A as in the computational tables.

Selector	n	Time (s) ↓	Mean LB	Mean UB	Status
Global risk	20	30.00±0.00	0.140	0.353	20T
Robust greedy	20	0.64±0.02	—	—	baseline
MMD screening	20	0.00±0.00	—	—	baseline
Additive MMD	20	0.02±0.00	—	—	baseline
L1 logistic	20	0.00±0.00	—	—	baseline
Random	20	0.00±0.00	—	—	baseline

Selection frequency asks a more interpretable question than showing twenty unrelated index lists: how often did a method choose a particular measurement across the prescribed splits? Figure 9 displays that descriptive frequency. The denominator is the recorded valid split outcomes for the relevant selector and dataset; failures and any difference from the planned denominator are explicitly recorded with the table.

A sparse frequency pattern can reflect several interchangeable correlated features. A diffuse pattern can reflect small selection samples or genuinely similar empirical subsets. These interpretations should be checked against prediction and solver gaps, rather than assuming that the most frequent feature is uniquely necessary. No synthetic-support recovery score is available here.

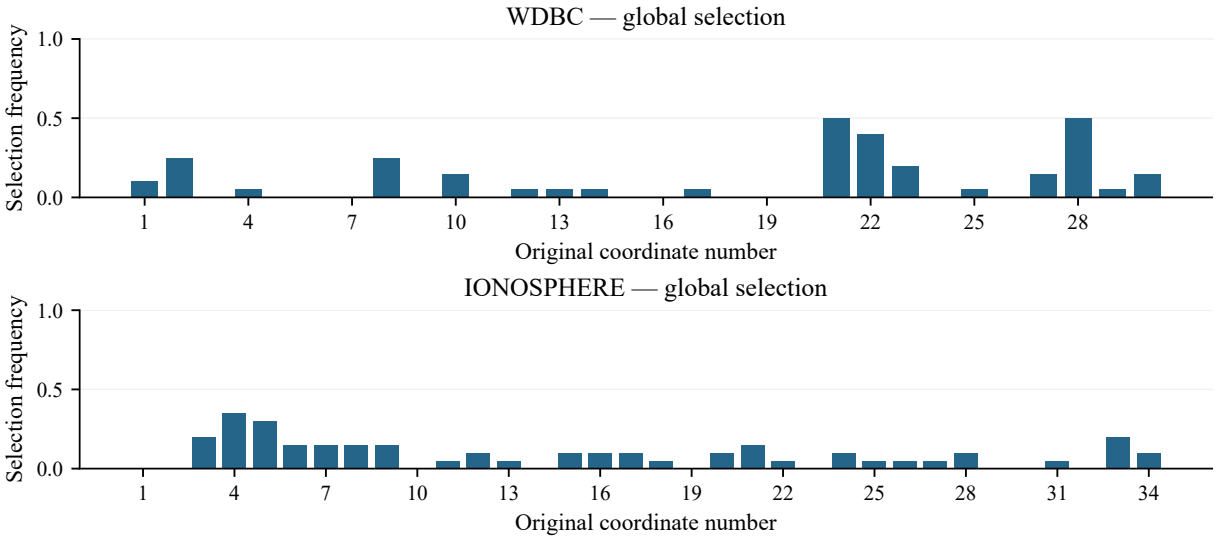


Figure 9: Feature selection frequency on the two public datasets across predeclared stratified splits. Frequency is descriptive split stability, not a causal importance estimate. Coordinates have fixed dataset identities. Counts use the recorded valid split outcomes, with failures and effective denominators disclosed in the accompanying tables. Prediction summaries use repeated-split mean and standard deviation, because the splits reuse records.

11 Exploratory supplement: retain the whole dual fraction

11.1 The experiment-motivated question

The primary timing audit exposed a weakness of the tangent master: it may repeatedly solve expensive integer models while retaining only a linear approximation to each useful dual fraction. This motivates a supplementary question: can the same certificate constrain future integer supports more effectively if its entire fraction is retained?

The supplement was declared separately after the primary computational observations and before its own runs. It keeps the risk objective, oracle, warm support, root stage, wall limit, and stopping rule. The treatment changes only the master representation, and its outcomes remain exploratory.

11.2 From a certificate to one linear master row

Store a repaired conic certificate with positive scale τ, nonnegative prices $\widehat{\mu}$, and a retained lower numerator $0 \le v \le \mathcal{H}(\widehat{\Gamma})$. At any support, its valid lower model is

$$f_c(z) = \frac{v}{\tau + \widehat{\mu}^\top z}, \qquad V(z) \ge f_c(z) \quad \text{for binary } z. \tag{13}$$

The denominator is positive because $\tau > 0$. Dividing the whole mass-and-price certificate by that denominator restores the oracle normalization at the new support. Homogeneity divides its objective by the same amount. This is why the stored fraction remains valid away from its generating support.

Let U be an independently feasible original-test upper bound. Restrict $0 \le \eta \le U$ and introduce a *shared* product vector $y_j = \eta z_j$. At binary masks its exact linear representation is

$$0 \le y_j \le U z_j, \qquad y_j \le \eta, \qquad y_j \ge \eta - U(1 - z_j). \tag{14}$$

Then $\eta \ge f_c(z)$ is equivalent to the linear row

$$v \le \tau\eta + \widehat{\mu}^\top y, \qquad \mathbf{1}^\top y = d\eta. \tag{15}$$

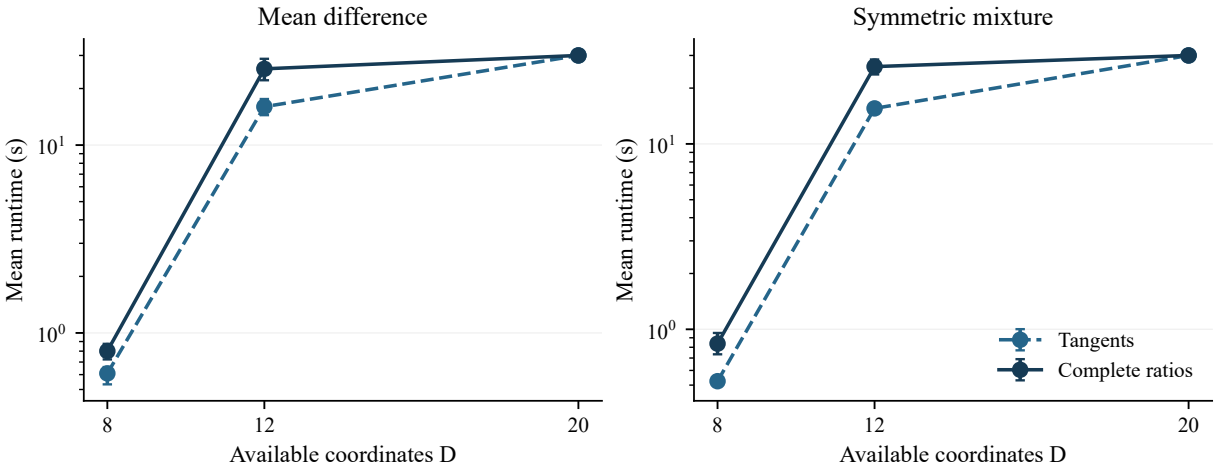


Figure 10: Exploratory paired master comparison on separately declared seeds. Both variants use the same robust objective, risk-gate oracle, initial support, five root rounds, and 30-second limit. The supplement retains full ratio rows and affine rows with one shared product vector. Uncertainty, where shown, resamples paired training instances; solved fraction includes time limits and failures. This post-primary supplement is reported separately and does not establish universal speedup.

The budget identity is valid because the search uses full-budget masks. All certificates share the same D product variables; each adds only one ratio row. An original optimal support with $\eta = \mathrm{OPT}$ and $y = \mathrm{OPT}\, z$ remains feasible, preserving the global lower/upper bound argument.

The master keeps the valid affine tangents as well. The product lift is exact at binary points, but a fractional linear-relaxation point need not satisfy $y = \eta z$. Therefore its projection need not dominate every affine tangent. A concrete counterexample is provided in the supplementary derivation. Retaining both row families prevents that information loss.

For an incumbent $U > 0$, the ratio also implies

$$\widehat{\mu}^{\top} z \geq v/U - \tau \quad \text{whenever } V(z) \leq U. \tag{16}$$

This says a competitive support must collect enough certificate price to reduce the retained overlap below the incumbent. It is an incumbent-dependent necessary condition, not a sufficient condition for a good support. It is already implied by the full ratio lift and bounds, so explicitly adding it is not claimed as a novel strengthening.

11.3 Paired results and their scope

The supplementary design uses mean and mixture data, eight selection observations per class, $(D, d) = (8, 2), (12, 3), (20, 3)$, seeds 10–14, five root rounds, no core-point selection, and the same 30-second limit. Each instance pairs the primary tangent master with the ratio-plus-tangent master. Total time includes its larger master and all unsuccessful outcomes.

Mean difference — supplementary readout. At $D = 8$, the observed mean ratio-minus-tangent charged time is +0.19 s over 5 paired seeds; complete ratios solve 5/5 recorded instances and tangents 5/5. At $D = 12$, the observed mean ratio-minus-tangent charged time is +9.45 s over 5 paired seeds; complete ratios solve 3/5 recorded instances and tangents 5/5. At $D = 20$, the observed mean ratio-minus-tangent charged time is +0.00 s over 5 paired seeds; complete ratios solve 0/5 recorded instances and tangents 0/5. Negative time differences favor the supplement on these paired instances. The table gives seed-level uncertainty and achieved gaps; no universal speed or superiority claim follows.

Table 21: Mean difference: paired exploratory master comparison. Time is charged mean seconds [95% seed-bootstrap interval], with all outcomes retained. Gap is mean absolute upper-minus-lower bound; visits count distinct integer supports. O: numerical optimum; T: time limit; M: master failure; F: oracle failure; A: accuracy limit.

Setting / method	Solved / n	Time (s) ↓	Gap ↓	Visits	Status
$D = 8, d = 2$ Global risk	5/5	0.61 [0.53, 0.66]	2.0×10^{-7}	27.8	5O
$D = 8, d = 2$ Complete ratios	5/5	0.80 [0.72, 0.88]	1.2×10^{-9}	27.4	5O
$D = 12, d = 3$ Global risk	5/5	16.01 [14.43, 17.59]	2.0×10^{-7}	197.8	5O
$D = 12, d = 3$ Complete ratios	3/5	25.46 [22.13, 28.79]	0.009	161.6	3O, 2T
$D = 20, d = 3$ Global risk	0/5	30.00 [30.00, 30.00]	0.115	178.8	5T
$D = 20, d = 3$ Complete ratios	0/5	30.00 [30.00, 30.00]	0.083	84.4	5T

Table 22: Within-seed ratio-minus-tangent charged time. Negative differences favor the supplement; these are descriptive paired intervals, not significance tests.

Dimension	Paired n	Δ time (s) [95% CI]
$D = 8$	5	0.19 [0.12, 0.25]
$D = 12$	5	9.45 [7.38, 11.39]
$D = 20$	5	0.00 [-0.00, 0.00]

Symmetric mixture — supplementary readout. At $D = 8$, the observed mean ratio-minus-tangent charged time is +0.31 s over 5 paired seeds; complete ratios solve 5/5 recorded instances and tangents 5/5. At $D = 12$, the observed mean ratio-minus-tangent charged time is +10.57 s over 5 paired seeds; complete ratios solve 4/5 recorded instances and tangents 5/5. At $D = 20$, the observed mean ratio-minus-tangent charged time is +0.00 s over 5 paired seeds; complete ratios solve 0/5 recorded instances and tangents 0/5. Negative time differences favor the supplement on these paired instances. The table gives seed-level uncertainty and achieved gaps; no universal speed or superiority claim follows.

Table 23: Symmetric mixture: paired exploratory master comparison. Time is charged mean seconds [95% seed-bootstrap interval], with all outcomes retained. Gap is mean absolute upper-minus-lower bound; visits count distinct integer supports. O: numerical optimum; T: time limit; M: master failure; F: oracle failure; A: accuracy limit.

Setting / method	Solved / n	Time (s) ↓	Gap ↓	Visits	Status
$D = 8, d = 2$ Global risk	5/5	0.52 [0.50, 0.56]	4.0×10^{-7}	27.2	5O
$D = 8, d = 2$ Complete ratios	5/5	0.84 [0.73, 0.96]	1.3×10^{-9}	27.0	5O
$D = 12, d = 3$ Global risk	5/5	15.54 [14.63, 16.59]	4.0×10^{-7}	192.0	5O
$D = 12, d = 3$ Complete ratios	4/5	26.11 [23.69, 28.54]	0.002	157.8	4O, 1T
$D = 20, d = 3$ Global risk	0/5	30.00 [30.00, 30.00]	0.101	179.6	5T

Continued on next page

Continued

Setting / method	Solved / n	Time (s) ↓	Gap ↓	Visits	Status
$D = 20, d = 3$		30.00			
Complete ratios	0/5	[30.00, 30.00]	0.072	83.2	5T

Table 24: Within-seed ratio-minus-tangent charged time. Negative differences favor the supplement; these are descriptive paired intervals, not significance tests.

Dimension	Paired n	Δ time (s) [95% CI]
$D = 8$	5	0.31 [0.22, 0.40]
$D = 12$	5	10.57 [7.36, 13.79]
$D = 20$	5	0.00 [0.00, 0.00]

These observations use seeds 10–14 in the separately frozen supplement. The tangent comparator is from this suite itself; primary-suite timings are not substituted. The component-time tables in the appendix identify whether a difference comes from master or oracle work.

The full ratio contains at least as much information as its own matched tangent at an integer support. Whether that helps total time depends on branch-and-bound behavior, numerical scaling, and the larger master. The comparison therefore shows both attained bounds and cost decomposition. Its scope is the declared instances and the computational representation of the same objective.

For numerical certificates, the retained numerator and actual normalization are recomputed after repair. The implementation keeps the valid ratio and affine rows, and omits zero-scale or zero-numerator certificates. Appendix B explains the matched-tangent comparison.

12 Reproducibility and conclusions supported by this study

The core implementation is independent of the historical COPT-based project. It separates data validation, convex oracles and deployment, and the outer global search. Experiment modules provide frozen generators, baselines, metrics, and a runner. Core, logistic, experiment, and plotting dependencies are declared separately in the new environment. Archived historical files are not imported by the new package or used as new results.

Each run preserves its configuration snapshot and digest, selected observed and semantic coordinates, preprocessing parameters, split and permutation seeds, solver bounds and status, elapsed timings, head certificate, stress evaluations, and raw rejection records. Oracle/master traces are separate. A resumed record is accepted only when its configuration digest matches. Third-party source references and data caches are excluded from publication of the independently written repository; source links and retrieval manifests preserve provenance.

The supported theoretical statement is an exact binary objective with a stronger problem-specific convex extension and globally valid lower cuts. The implementation's numerical evidence checks this against independent small transport problems and deployment constraints. The main empirical conclusion must reflect the full result tables: optimization exactness, practical search efficiency, and independent statistical utility can diverge. Unfavorable computational and statistical outcomes remain part of that conclusion.

The experiment sizes are intentionally controlled. The computational design reaches moderate feature dimensions with small source pools, and the statistical design uses small selection samples. Timeouts prevent treating every high-dimensional incumbent as exact. Public-data repeated splits do not supply new independent populations. Fixed radii and baseline hyperparameters make the comparison reproducible but do not establish optimal tuning. The exploratory ratio master addresses one observed computational limitation and retains its separate status.

Further work should be tied to these findings: improve the master representation or callback implementation if master cost dominates; study objective/radius selection on independent development data if empirical risk and power disagree; and increase selection samples when apparent support choices are unstable. Those are questions raised by the records, not completed claims of this report.

A Complete settings and aggregation rules

A.1 Settings that define each comparison

Table 25: Predeclared settings. Counts describe the planned design; completed outcome counts and failures are supplied by the final result inventory.

Experiment	Settings
Primary statistical	Four mechanisms; $D = 12$, signal size and budget three; seeds 0–29; selection and downstream samples 20 per class; classification 1,000 per class; twenty null and twenty alternative batches per seed, each 40 per class; 199 permutations; equal radii 0.25.
Small computation	Mean and mixture; $(D, d) = (8, 2), (12, 3)$; eight source atoms per class; seeds 0–9; box, triangle, risk without roots, risk with five root rounds, and enumeration.
Scaling computation	Mean and mixture; $D = 20, 40, 80$, $d = 3$; ten atoms per class; seeds 0–4; the four gate/root variants.
Real data	WDBC and Ionosphere; seeds 0–19; selection and downstream samples 20 per class with disjoint indices; remaining records for classification; budget three; equal radii 0.25.
Radius sensitivity	Mean and mixture; seeds 0–9; primary statistical sample sizes; risk selector at $0.10, 0.25, 0.50$; no final-data tuning.
Ratio supplement	Mean and mixture; seeds 10–14; eight source atoms per class; $(D, d) = (8, 2), (12, 3), (20, 3)$; tangent versus ratio-plus-tangent, five root rounds, no core-point selection.
Common controls	Thirty-second selection budget where optimization is used; absolute tolerance 10^{-6} and relative tolerance 10^{-5}; fixed shared initial support; single solver/BLAS thread; common data, scaling, and evaluation within paired seeds.

Synthetic uncertainty is computed from one aggregate per independently trained seed. For power, average the twenty conditional rejection indicators first; for prediction, compute one class-balanced test value; for recovery, compute one support value. Bootstrap these seed-level objects with 2,000 fixed-seed resamples and percentile endpoints. Preserve paired seed alignment for method differences. A missing metric retains its effective denominator rather than being silently filled with zero.

Real-data means and standard deviations use valid split outcomes and disclose their count. Because split indices overlap across runs, these are descriptive repeated-split summaries. Solver solved fraction uses all scheduled outcomes, not only rows with completed prediction metrics. Any early failure has its failure record, timing, and available feasible incumbent retained.

A.2 Names and normalization in the saved records

The same factor-of-two convention applies to robust risk and randomized balanced error. A half-hinge surrogate average and randomized balanced error should not appear as two independent wins in a combined score. Likewise, class-zero prediction error is not the null rejection rate of an equality test. The main figures separate those endpoints.

Table 26: Saved metric names and their meaning. Prefixes identify the common robust or support-vector head.

Record field	Meaning
`selection_lower`, `selection_upper`	Global lower bound and feasible original-test upper bound for selection.
`selection_gap`	Upper minus lower; optimization uncertainty.
`power`, `null_rejection`	Seed-level alternative and null rejection fractions for the common equality test.
`fdp`, `ndp`, `f1`	Designated-signal recovery on synthetic data.
`randomized_balanced_error`	Mean of the two randomized class errors; same value as the half-hinge surrogate average.
`balanced_error`	Mean of classwise hard-decision errors at the declared threshold.
`auc`	Area under the ROC curve using class-one scores.
`predictive_log_loss_sum`	Sum of classwise mean negative log probabilities; its average is half the sum.
`empirical_certificate_violation`	Positive part of shifted-source surrogate sum minus the fitted robust upper bound.

B Additional algebra for the reproduced baseline and supplement

For Wang's paired sample construction, let $h_{il,j} = k_j(x_i, x_l) + k_j(y_i, y_l) - k_j(x_i, y_l) - k_j(y_i, x_l)$. The mean and inspected row-based variance coefficients are

$$t = \frac{1}{n(n-1)} \sum_{i \neq l} h_{il}, \qquad r_i = \frac{1}{n} \sum_{l} h_{il}, \qquad \widehat{\Sigma} = \frac{4}{n} \sum_{i} (r_i - \bar{r})(r_i - \bar{r})^\top. \tag{17}$$

The paper criterion uses $A = -0.1\widehat{\Sigma}$ in $t^\top \omega + \omega^\top A \omega$. Independent scalar-kernel fixtures check these quantities and also preserve the inspected source's positive sign. On a fixed support, diagonalize A; stationarity gives $x_i = (t_i/2)/(\lambda - d_i)$. Solve the norm equation above the top eigenvalue, with a separate top-eigenspace fill for the hard case. This provides a global weight fit, while support enumeration or labeled approximate search handles sparsity.

For the ratio supplement, write $D_c(z) = \tau + \widehat{\mu}^\top z$ and $\nu = D_c(z^t)$. The matched tangent of $v/D_c(z)$ is

$$T_c(z) = \frac{v}{\nu} - \frac{v}{\nu^2} \widehat{\mu}^\top (z - z^t), \qquad f_c(z) - T_c(z) = \frac{v[D_c(z) - \nu]^2}{D_c(z)\nu^2} \geq 0. \tag{18}$$

This precisely measures the curvature discarded by one tangent. It is an integer-support information comparison before product relaxation. It does not prove that a fractional product lift dominates the tangent, which is why both rows are retained.

One counterexample uses $f(z) = 1/(1 + 4z_1 + 5z_2 + z_3)$, budget one, anchor $(1, 0, 0)$, and $U = 1/5$. The fractional point $z = (0, 5/6, 1/6)$, $\eta = 1/6$, $y = (0, 1/6, 0)$ satisfies the product-relaxation inequalities, $\mathbf{1}^\top y = \eta$, and the ratio row at equality. But its $\eta = 1/6$ is below the matched tangent $14/75$. The common full-budget identity therefore helps without replacing the affine cuts.

C Code map and an inspectable route through one run

Table 27: The new implementation separates mathematical roles. Paths are relative to the repository root.

Source	Mathematical or experimental responsibility
`code/robusthw/problem.py`	Validate arrays, source weights, radii and budget; keep pooled atoms and coordinate differences; serialize cuts and result objects.
`code/robusthw/oracle.py`	Assemble fixed-support and fractional gate problems; solve LP/cone oracles; repair primal and dual certificates; construct deployable whole-space tests.
`code/robusthw/solver.py`	Manage wall budget, root and integer master calls, cuts, best feasible support, global bounds, retries, and stopping/accuracy statuses.
`experiments/data.py`	Independent generators and real-data indices; selection-only standardization; semantic coordinate maps.
`experiments/baselines.py`	Fully refitted comparators and independent paper-objective coefficient/weight reconstruction.
`experiments/metrics.py`	Unbiased statistics, permutation calibration, recovery, class-wise errors, ranking, and shifts.
`experiments/runner.py`	Frozen jobs, common evaluators, configuration digests, per-trial records, failures, and safe resumption.
`code/tests/`	Independent objective/cut/deployment checks and baseline/split/metric checks.
`configs/`, `scripts/`	Frozen design settings; environment/data setup; aggregation, figures, and report generation.

To inspect one result, begin with its configuration digest and data metadata. Confirm the selected support and its semantic mapping. Follow the oracle trace: a repaired primal supplies the feasible upper bound; a repaired dual supplies the lower value and cut. Follow the master trace: its global solver bound updates the global lower bound. Finally inspect the separately fitted head and the fresh evaluation records. This route exposes the relevant mathematical relationships without reading every utility function first.

The repository documentation explains environment setup and exact commands. The source and result inventories preserve the run version, hardware, configuration hashes, solver versions, and outcome counts. A final report must be regenerated from these recorded artifacts, rather than manually editing displayed means after a run changes.

D Full result tables and outcome inventory

The main report uses compact tables and figures to preserve the narrative. The following insertion contains complete settings-level summaries, effective sample counts, achieved bound/status information, and any unavailable entries. A dash denotes an unavailable or inapplicable quantity, not a successful zero. Failures and time limits remain explicit. Synthetic table intervals use training-seed bootstrap uncertainty; real-data table variability uses repeated-split standard deviation.

D.1 Component time and effort

Table 28: computational, Mean difference: all-outcome mean component times, calls, root bounds, and feasible original risk. These means include unsuccessful outcomes; an absent component is a dash.

Setting / method	n	Oracle (s)	Master (s)	Calls	Root LB	Risk UB
$D = 8$ Box	10	0.17	0.13	29.1	0.111	0.506
$D = 8$ Triangle	10	0.17	0.14	29.1	0.111	0.506
$D = 8$ Risk, no root	10	0.17	0.19	28.0	0.000	0.506
$D = 8$ Global risk	10	0.24	0.45	38.0	0.169	0.506
$D = 8$ Enumeration	10	—	—	28.0	—	0.506
$D = 12$ Box	10	1.54	2.51	243.4	0.063	0.273
$D = 12$ Triangle	10	1.40	3.21	221.3	0.063	0.273
$D = 12$ Risk, no root	10	1.36	12.35	203.2	0.000	0.273
$D = 12$ Global risk	10	1.36	15.16	202.1	0.111	0.273
$D = 12$ Enumeration	10	—	—	220.0	—	0.273
$D = 20$ Box	5	7.10	22.85	809.4	0.038	0.314
$D = 20$ Triangle	5	7.14	22.81	813.4	0.038	0.317
$D = 20$ Risk, no root	5	2.74	27.24	306.4	0.000	0.311
$D = 20$ Global risk	5	1.66	28.32	182.8	0.108	0.311
$D = 40$ Box	5	7.20	22.77	654.0	0.015	0.302
$D = 40$ Triangle	5	7.62	22.34	694.6	0.015	0.302
$D = 40$ Risk, no root	5	6.30	23.69	584.0	0.000	0.292
$D = 40$ Global risk	5	1.45	28.55	127.0	0.092	0.310
$D = 80$ Box	5	6.22	23.76	407.4	0.007	0.290
$D = 80$ Triangle	5	6.77	23.20	448.4	0.007	0.306
$D = 80$ Risk, no root	5	6.86	23.15	458.2	0.000	0.279
$D = 80$ Global risk	5	1.60	28.40	94.2	0.091	0.281

Table 29: computational, Symmetric mixture: all-outcome mean component times, calls, root bounds, and feasible original risk. These means include unsuccessful outcomes; an absent component is a dash.

Setting / method	n	Oracle (s)	Master (s)	Calls	Root LB	Risk UB
$D = 8$ Box	10	0.18	0.14	29.3	0.114	0.477
$D = 8$ Triangle	10	0.17	0.14	29.3	0.114	0.477

Continued on next page

Continued

Setting / method	n	Oracle (s)	Master (s)	Calls	Root LB	Risk UB
$D = 8$ Risk, no root	10	0.17	0.18	28.0	0.000	0.477
$D = 8$ Global risk	10	0.23	0.44	37.0	0.176	0.477
$D = 8$ Enumeration	10	—	—	28.0	—	0.477
$D = 12$ Box	10	1.39	2.11	221.0	0.062	0.275
$D = 12$ Triangle	10	1.40	3.34	220.2	0.062	0.275
$D = 12$ Risk, no root	10	1.32	11.88	200.0	0.000	0.275
$D = 12$ Global risk	10	1.35	15.02	201.2	0.111	0.275
$D = 12$ Enumeration	10	—	—	220.0	—	0.275
$D = 20$ Box	5	7.03	22.91	805.0	0.035	0.298
$D = 20$ Triangle	5	7.04	22.93	807.0	0.035	0.291
$D = 20$ Risk, no root	5	2.68	27.30	301.8	0.000	0.299
$D = 20$ Global risk	5	1.74	28.24	192.8	0.106	0.295
$D = 40$ Box	5	7.25	22.71	656.8	0.015	0.278
$D = 40$ Triangle	5	7.57	22.39	693.6	0.015	0.298
$D = 40$ Risk, no root	5	6.28	23.71	582.2	0.000	0.268
$D = 40$ Global risk	5	1.48	28.51	131.4	0.093	0.287
$D = 80$ Box	5	6.30	23.67	420.2	0.007	0.293
$D = 80$ Triangle	5	6.94	23.04	461.4	0.007	0.295
$D = 80$ Risk, no root	5	6.51	23.48	442.2	0.000	0.291
$D = 80$ Global risk	5	1.67	28.32	100.8	0.093	0.293

Table 30: ratio, Mean difference: all-outcome mean component times, calls, root bounds, and feasible original risk. These means include unsuccessful outcomes; an absent component is a dash.

Setting / method	n	Oracle (s)	Master (s)	Calls	Root LB	Risk UB
$D = 8$ Global risk	5	0.20	0.41	32.4	0.180	0.477
$D = 8$ Complete ratios	5	0.20	0.60	32.0	0.180	0.477
$D = 12$ Global risk	5	1.40	14.58	202.0	0.113	0.297
$D = 12$ Complete ratios	5	1.20	24.25	165.8	0.113	0.297
$D = 20$ Global risk	5	1.35	28.63	183.0	0.096	0.257
$D = 20$ Complete ratios	5	0.71	29.28	88.6	0.096	0.256

Table 31: ratio, Symmetric mixture: all-outcome mean component times, calls, root bounds, and feasible original risk. These means include unsuccessful outcomes; an absent component is a dash.

Setting / method	n	Oracle (s)	Master (s)	Calls	Root LB	Risk UB
$D = 8$ Global risk	5	0.20	0.33	31.2	0.179	0.470
$D = 8$ Complete ratios	5	0.20	0.64	31.0	0.179	0.470
$D = 12$ Global risk	5	1.37	14.15	196.6	0.122	0.280
$D = 12$ Complete ratios	5	1.15	24.94	162.4	0.122	0.280
$D = 20$ Global risk	5	1.34	28.64	183.8	0.092	0.243
$D = 20$ Complete ratios	5	0.67	29.32	87.4	0.092	0.244

D.2 Diagnostic supports and probability loss

Table 32: Reference supports are diagnostic, not six-method budget-matched competitors. Known signal uses ground truth; all features exceed the acquisition budget. Values are means [95% training-seed intervals].

Diagnostic reference	n	Power ↑	Null rejection	Random BER ↓
Mean difference Known signal	30	1.000 [1.000, 1.000]	0.053 [0.038, 0.070]	0.414 [0.405, 0.422]
Mean difference All features	30	0.998 [0.995, 1.000]	0.055 [0.037, 0.073]	0.497 [0.496, 0.497]
Variance difference Known signal	30	0.957 [0.940, 0.973]	0.050 [0.035, 0.065]	0.445 [0.441, 0.450]
Variance difference All features	30	0.138 [0.112, 0.167]	0.057 [0.038, 0.077]	0.498 [0.497, 0.498]
Equal marginals, different dependence Known signal	30	0.218 [0.187, 0.250]	0.055 [0.040, 0.070]	0.471 [0.467, 0.474]
Equal marginals, different dependence All features	30	0.060 [0.045, 0.077]	0.050 [0.033, 0.068]	0.499 [0.499, 0.499]
Symmetric mixture Known signal	30	1.000 [1.000, 1.000]	0.048 [0.032, 0.065]	0.351 [0.342, 0.359]
Symmetric mixture All features	30	0.207 [0.163, 0.253]	0.058 [0.038, 0.080]	0.493 [0.492, 0.494]

Table 33: statistical: predictive cross-entropy of the half-hinge-trained common head, semantic support stability, and charged selection time. Synthetic intervals are seed-bootstrap intervals; real entries use split SD. Stability is pairwise Jaccard, not a confidence interval.

Dataset / selector	Valid n	Log loss ↓	Stability ↑	Time (s) ↓
mean Global risk	30	0.670 [0.653, 0.687]	0.246	21.15 [20.72, 21.56]
mean Robust greedy	30	0.666 [0.649, 0.681]	0.276	0.21 [0.20, 0.21]
mean MMD screening	30	0.642 [0.627, 0.657]	0.452	0.00 [0.00, 0.00]
mean Additive MMD	30	0.651 [0.633, 0.670]	0.430	0.04 [0.04, 0.04]
mean L1 logistic	30	0.645 [0.629, 0.661]	0.414	0.00 [0.00, 0.00]

Continued on next page

Continued

Dataset / selector	Valid n	Log loss ↓	Stability ↑	Time (s) ↓
mean		0.729		0.00
Random	30	[0.714, 0.745]	0.157	[0.00, 0.00]
variance		0.725		22.11
Global risk	30	[0.714, 0.737]	0.200	[21.16, 23.26]
variance		0.722		0.21
Robust greedy	30	[0.711, 0.733]	0.183	[0.21, 0.21]
variance		0.714		0.00
MMD screening	30	[0.703, 0.725]	0.248	[0.00, 0.00]
variance		0.723		0.04
Additive MMD	30	[0.712, 0.735]	0.209	[0.04, 0.04]
variance		0.737		0.00
L1 logistic	30	[0.727, 0.748]	0.179	[0.00, 0.00]
variance		0.744		0.00
Random	30	[0.734, 0.753]	0.157	[0.00, 0.00]
dependence		0.759		21.32
Global risk	30	[0.751, 0.766]	0.164	[20.94, 21.71]
dependence		0.762		0.21
Robust greedy	30	[0.755, 0.769]	0.164	[0.20, 0.21]
dependence		0.757		0.00
MMD screening	30	[0.750, 0.764]	0.169	[0.00, 0.00]
dependence		0.758		0.04
Additive MMD	30	[0.751, 0.764]	0.156	[0.04, 0.04]
dependence		0.761		0.00
L1 logistic	30	[0.753, 0.769]	0.167	[0.00, 0.00]
dependence		0.763		0.00
Random	30	[0.757, 0.770]	0.157	[0.00, 0.00]
mixture		0.614		22.24
Global risk	30	[0.590, 0.640]	0.316	[21.33, 23.31]
mixture		0.594		0.21
Robust greedy	30	[0.567, 0.623]	0.372	[0.21, 0.22]
mixture		0.588		0.00
MMD screening	30	[0.561, 0.614]	0.403	[0.00, 0.00]
mixture		0.606		0.04
Additive MMD	30	[0.581, 0.629]	0.338	[0.04, 0.04]
mixture		0.720		0.00
L1 logistic	30	[0.699, 0.738]	0.186	[0.00, 0.00]
mixture		0.723		0.00
Random	30	[0.702, 0.742]	0.157	[0.00, 0.00]

Table 34: real: predictive cross-entropy of the half-hinge-trained common head, semantic support stability, and charged selection time. Synthetic intervals are seed-bootstrap intervals; real entries use split SD. Stability is pairwise Jaccard, not a confidence interval.

Dataset / selector	Valid n	Log loss ↓	Stability ↑	Time (s) ↓
wdbc				
Global risk	20	0.329±0.059	0.179	30.00±0.00
wdbc				
Robust greedy	20	0.291±0.039	0.233	0.54±0.02
wdbc				
MMD screening	20	0.255±0.030	0.398	0.00±0.00
wdbc				
Additive MMD	20	0.252±0.029	0.387	0.02±0.00
wdbc				
L1 logistic	20	0.302±0.047	0.266	0.00±0.00
wdbc				
Random	20	0.431±0.105	0.062	0.00±0.00

Continued on next page

Continued

Dataset / selector	Valid n	Log loss ↓	Stability ↑	Time (s) ↓
ionosphere Global risk	20	0.480±0.079	0.082	30.00±0.00
ionosphere Robust greedy	20	0.513±0.232	0.153	0.64±0.02
ionosphere MMD screening	20	0.828±0.567	0.455	0.00±0.00
ionosphere Additive MMD	20	0.753±0.527	0.452	0.02±0.00
ionosphere L1 logistic	20	0.471±0.094	0.141	0.00±0.00
ionosphere Random	20	0.553±0.096	0.058	0.00±0.00

Table 35: Wang paper-objective weights with the independently frozen native additive statistic. This diagnostic evaluates the learned weighted statistic, rather than the common joint-RBF support test. Means [95% seed intervals].

Mechanism	Valid n	Native power ↑	Native null rejection
Mean difference	30	0.977 [0.950, 0.998]	0.045 [0.025, 0.065]
Variance difference	30	0.518 [0.408, 0.625]	0.052 [0.033, 0.072]
Equal marginals, different dependence	30	0.045 [0.030, 0.060]	0.052 [0.035, 0.070]
Symmetric mixture	30	0.945 [0.915, 0.970]	0.040 [0.025, 0.057]

D.3 Effective denominators, failures, and source checks

Table 36: Every incomplete evaluation or nonoptimal certifying selector group. Evaluation completion and numerical optimality are distinct; status counts use all recorded trials.

Setting	Present	Evaluated	Final selection status counts
computational, mean $D = 8, \theta = 0.25$; Global risk	10	10	1 master failure, 9 optimal
computational, mean $D = 20, \theta = 0.25$; Box	5	5	5 time limit
computational, mean $D = 20, \theta = 0.25$; Global risk	5	5	5 time limit
computational, mean $D = 20, \theta = 0.25$; Risk, no root	5	5	5 time limit
computational, mean $D = 20, \theta = 0.25$; Triangle	5	5	5 time limit
computational, mean $D = 40, \theta = 0.25$; Box	5	5	5 time limit
computational, mean $D = 40, \theta = 0.25$; Global risk	5	5	5 time limit
computational, mean $D = 40, \theta = 0.25$; Risk, no root	5	5	5 time limit
computational, mean $D = 40, \theta = 0.25$; Triangle	5	5	5 time limit
computational, mean $D = 80, \theta = 0.25$; Box	5	5	5 time limit

Continued on next page

Continued

Setting	Present	Evaluated	Final selection status counts
computational, mean $D = 80, \theta = 0.25$; Global risk	5	5	5 time limit
computational, mean $D = 80, \theta = 0.25$; Risk, no root	5	5	5 time limit
computational, mean $D = 80, \theta = 0.25$; Triangle	5	5	5 time limit
computational, mixture $D = 20, \theta = 0.25$; Box	5	5	5 time limit
computational, mixture $D = 20, \theta = 0.25$; Global risk	5	5	5 time limit
computational, mixture $D = 20, \theta = 0.25$; Risk, no root	5	5	5 time limit
computational, mixture $D = 20, \theta = 0.25$; Triangle	5	5	5 time limit
computational, mixture $D = 40, \theta = 0.25$; Box	5	5	5 time limit
computational, mixture $D = 40, \theta = 0.25$; Global risk	5	5	5 time limit
computational, mixture $D = 40, \theta = 0.25$; Risk, no root	5	5	5 time limit
computational, mixture $D = 40, \theta = 0.25$; Triangle	5	5	5 time limit
computational, mixture $D = 80, \theta = 0.25$; Box	5	5	5 time limit
computational, mixture $D = 80, \theta = 0.25$; Global risk	5	5	5 time limit
computational, mixture $D = 80, \theta = 0.25$; Risk, no root	5	5	5 time limit
computational, mixture $D = 80, \theta = 0.25$; Triangle	5	5	5 time limit
ratio, mean $D = 12, \theta = 0.25$; Complete ratios	5	5	3 optimal, 2 time limit
ratio, mean $D = 20, \theta = 0.25$; Global risk	5	5	5 time limit
ratio, mean $D = 20, \theta = 0.25$; Complete ratios	5	5	5 time limit
ratio, mixture $D = 12, \theta = 0.25$; Complete ratios	5	5	4 optimal, 1 time limit
ratio, mixture $D = 20, \theta = 0.25$; Global risk	5	5	5 time limit
ratio, mixture $D = 20, \theta = 0.25$; Complete ratios	5	5	5 time limit
real, ionosphere $D = 34, \theta = 0.25$; Global risk	20	20	20 time limit
real, wdbc $D = 30, \theta = 0.25$; Global risk	20	20	20 time limit
sensitivity, mixture $D = 12, \theta = 0.25$; Global risk	10	10	1 master failure, 9 optimal
sensitivity, mixture $D = 12, \theta = 0.5$; Global risk	10	10	1 master failure, 9 optimal
statistical, mixture $D = 12, \theta = 0.25$; Global risk	30	30	27 optimal, 3 time limit
statistical, variance $D = 12, \theta = 0.25$; Global risk	30	30	27 optimal, 3 time limit

Table 37: Effective metric denominators. Test and head counts are numbers of training seeds/splits with valid seed-level metrics; conditional test batches are not extra selections. A dash denotes an absent result and n/a an undefined real-data power target.

Setting / selector	Present	Test n	Head n	Source-check n
statistical, dependence $\theta = 0.25$; All features	30	30	30	30
statistical, dependence $\theta = 0.25$; Robust greedy	30	30	30	30
statistical, dependence $\theta = 0.25$; L1 logistic	30	30	30	30
statistical, dependence $\theta = 0.25$; MMD screening	30	30	30	30
statistical, dependence $\theta = 0.25$; Random	30	30	30	30
statistical, dependence $\theta = 0.25$; Global risk	30	30	30	30
statistical, dependence $\theta = 0.25$; Known signal	30	30	30	30
statistical, dependence $\theta = 0.25$; Additive MMD	30	30	30	30
statistical, mean $\theta = 0.25$; All features	30	30	30	30
statistical, mean $\theta = 0.25$; Robust greedy	30	30	30	30
statistical, mean $\theta = 0.25$; L1 logistic	30	30	30	30
statistical, mean $\theta = 0.25$; MMD screening	30	30	30	30
statistical, mean $\theta = 0.25$; Random	30	30	30	30
statistical, mean $\theta = 0.25$; Global risk	30	30	30	30
statistical, mean $\theta = 0.25$; Known signal	30	30	30	30
statistical, mean $\theta = 0.25$; Additive MMD	30	30	30	30
statistical, mixture $\theta = 0.25$; All features	30	30	30	30
statistical, mixture $\theta = 0.25$; Robust greedy	30	30	30	30
statistical, mixture $\theta = 0.25$; L1 logistic	30	30	30	30
statistical, mixture $\theta = 0.25$; MMD screening	30	30	30	30
statistical, mixture $\theta = 0.25$; Random	30	30	30	30
statistical, mixture $\theta = 0.25$; Global risk	30	30	30	30
statistical, mixture $\theta = 0.25$; Known signal	30	30	30	30
statistical, mixture $\theta = 0.25$; Additive MMD	30	30	30	30
statistical, variance $\theta = 0.25$; All features	30	30	30	30
statistical, variance $\theta = 0.25$; Robust greedy	30	30	30	30
statistical, variance $\theta = 0.25$; L1 logistic	30	30	30	30

Continued on next page

Continued

Setting / selector	Present	Test n	Head n	Source-check n
statistical, variance $\theta = 0.25$; MMD screening	30	30	30	30
statistical, variance $\theta = 0.25$; Random	30	30	30	30
statistical, variance $\theta = 0.25$; Global risk	30	30	30	30
statistical, variance $\theta = 0.25$; Known signal	30	30	30	30
statistical, variance $\theta = 0.25$; Additive MMD	30	30	30	30
real, ionosphere $\theta = 0.25$; Robust greedy	20	n/a	20	20
real, ionosphere $\theta = 0.25$; L1 logistic	20	n/a	20	20
real, ionosphere $\theta = 0.25$; MMD screening	20	n/a	20	20
real, ionosphere $\theta = 0.25$; Random	20	n/a	20	20
real, ionosphere $\theta = 0.25$; Global risk	20	n/a	20	20
real, ionosphere $\theta = 0.25$; Additive MMD	20	n/a	20	20
real, wdbc $\theta = 0.25$; Robust greedy	20	n/a	20	20
real, wdbc $\theta = 0.25$; L1 logistic	20	n/a	20	20
real, wdbc $\theta = 0.25$; MMD screening	20	n/a	20	20
real, wdbc $\theta = 0.25$; Random	20	n/a	20	20
real, wdbc $\theta = 0.25$; Global risk	20	n/a	20	20
real, wdbc $\theta = 0.25$; Additive MMD	20	n/a	20	20
sensitivity, mean $\theta = 0.1$; Global risk	10	10	10	10
sensitivity, mean $\theta = 0.25$; Global risk	10	10	10	10
sensitivity, mean $\theta = 0.5$; Global risk	10	10	10	10
sensitivity, mixture $\theta = 0.1$; Global risk	10	10	10	10
sensitivity, mixture $\theta = 0.25$; Global risk	10	10	10	10
sensitivity, mixture $\theta = 0.5$; Global risk	10	10	10	10

There are 1260 explicit shifted-source certificate checks. The largest recorded nonnegative surrogate violation is 0.000; 0 exceed 10^{-7}. This directly budget-feasible source-atom calculation is separate from population stress.

Table 38: Preserved failed attempts and their final records. Final time is cumulative charged selection time, not a replacement that erases the original attempt.

Trial	Original status	Final status	Original (s)	Charged (s)
computational, mean $D = 8$, seed 3; Global risk	master failure	optimal	0.62	1.31
computational, mean $D = 8$, seed 9; Global risk	master failure	master failure	0.44	0.97
computational, mean $D = 12$, seed 3; Box	master failure	optimal	3.94	7.50
computational, mixture $D = 8$, seed 8; Global risk	master failure	optimal	0.66	1.33
computational, mixture $D = 8$, seed 9; Global risk	master failure	optimal	0.54	1.14
statistical, mean $D = 12$, seed 3; Global risk	master failure	optimal	1.25	23.36
statistical, mixture $D = 12$, seed 1; Global risk	master failure	time limit	20.27	30.00
statistical, mixture $D = 12$, seed 6; Global risk	master failure	time limit	20.92	30.00
statistical, mixture $D = 12$, seed 13; Global risk	master failure	time limit	13.85	30.00
statistical, variance $D = 12$, seed 4; Global risk	master failure	time limit	19.54	30.00
statistical, variance $D = 12$, seed 8; Global risk	master failure	time limit	19.92	30.00
statistical, variance $D = 12$, seed 21; Global risk	master failure	time limit	10.51	30.00

E Source provenance and reproduction boundaries

The local reading records identify the PDF versions and relevant sections. Wang's local paper is arXiv:2302.07415v4, including its mathematical supplement; Xie's local robust-testing source is arXiv:2105.14348v1; Gao's is arXiv:1805.10611v1; Swaroop's is arXiv:2510.21113v1. These records distinguish stated parameters from details not supplied by the source.

The author-linked Wang repository was inspected at the following commit:

`031f740e9ca6672751145b16de62cfaf731e98de`.

The revision coefficient and optimizer files are read-only references. Our independently written criterion and evaluation adaptations are stated in Section 5 and `docs/baseline_reproduction.md`; third-party unlicensed source is not redistributed. The robust-testing source archives contain paper sources rather than a located executable experiment repository, so no exact replay of their reported tables is claimed.

The real-data manifest stores original download URLs, persistent DOIs, CC BY 4.0 licensing, SHA-256 hashes, dimensions, label mappings, and excluded columns. Raw records and external paper/code caches are not required as redistributed repository contents when their setup script can recover the public sources. The publication artifact is the new implementation, frozen configurations, documentation, and reproducible run records with their provenance.

References

[1] D. Bertsimas, R. Cory-Wright, and J. Pauphilet. *A Unified Approach to Mixed-Integer Optimization Problems with Logical Constraints*. Local source arXiv:1907.02109v3. Primary paper.

[2] Liyan Xie, Rui Gao, and Yao Xie. *Robust Hypothesis Testing with Wasserstein Uncertainty Sets*. 2021. arXiv:2105.14348.

[3] Rui Gao, Liyan Xie, Yao Xie, and Huan Xu. *Robust Hypothesis Testing Using Wasserstein Uncertainty Sets*. 2018. arXiv:1805.10611.

[4] Jie Wang, Santanu S. Dey, and Yao Xie. *Variable Selection for Kernel Two-Sample Tests*. Local source arXiv:2302.07415v4, 2024. Primary paper.

[5] Author-linked implementation for Wang et al. WalterBabyRudin/MMDVar_Selection, inspected commit stated above.

[6] Maitreyi Swaroop, Tamar Krishnamurti, and Bryan Wilder. *Distributionally Robust Feature Selection*. NeurIPS 2025. Primary paper; Official implementation.

[7] Hiroyuki Hanada, Satoshi Akahane, Noriaki Hashimoto, Shion Takeno, and Ichiro Takeuchi. *Safe Distributionally Robust Feature Selection under Covariate Shift*. 2026. Primary paper.

[8] UCI Machine Learning Repository. *Breast Cancer Wisconsin (Diagnostic)*. Dataset DOI: 10.24432/C5DW2B.

[9] UCI Machine Learning Repository. *Ionosphere*. Dataset DOI: 10.24432/C5W01B.